

CO_2 and CH_4 fluxes in thawing permafrost in Finnmark, Norway

Faculty of Biology, Chemistry & Earth Sciences

Soil Ecology Group

Prof. Dr. Werner Borken

University Bayreuth

Master Thesis in Environmental Chemistry

Submitted by:

Ricardo Steven Soler Salgado

1. **Examiner:** Prof. Dr. Werner Borken
2. **Examiner:** Dr. Cristian Estop-Aragonés

Declaration of Authorship

I herewith certify that the work presented in the following with the title

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Is to the best of my knowledge and belief original and the result of my own investigations. Third party work, non-published and published information I received are properly and duly acknowledged.

This work has never previously been submitted to any other examination committee.

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Abstract

Permafrost thaw significantly impacts carbon dynamics in Arctic peatlands by altering methane (CH_4) and carbon dioxide (CO_2) fluxes, with potential climate feedback effects. This study examined greenhouse gas (GHG) fluxes in palsa and thermokarst features at Iškoras and Áidejávri, Finnmark, Norway, over the course of the growing season from early Jule until early September. Using an in-situ closed chamber method, the study assessed the effects of permafrost degradation and accompanying abiotic variables on CH_4 and CO_2 fluxes, aiming to quantify GHG flux variability across thawing permafrost features. Results revealed that CH_4 fluxes were significantly higher in thermokarsts, with mean fluxes of 100.51 mg CH_4 m^{-2} d^{-1} in TR, 20.20 mg CH_4 m^{-2} d^{-1} in TM, 6.82 mg CH_4 m^{-2} d^{-1} in T, and 0.91 mg CH_4 m^{-2} d^{-1} in TY, while palsas acted as CH_4 sinks (mean: -3.19 mg CH_4 m^{-2} d^{-1}). The highest CH_4 fluxes were recorded mid-season, corresponding to warmer temperatures and higher microbial activity. CO_2 fluxes exhibited a gradual seasonal increase, peaking between mid and late July. TM recorded the highest mean CO_2 flux (5.72 g CO_2 m^{-2} d^{-1}), followed by T (5.16 g CO_2 m^{-2} d^{-1}), TY (5.02 g CO_2 m^{-2} d^{-1}), and TR (3.55 g CO_2 m^{-2} d^{-1}), while palsas showed an overall mean flux of 5.04 g CO_2 m^{-2} d^{-1}. Environmental drivers played distinct roles in regulating CH_4 and CO_2 fluxes. CH_4 fluxes in thermokarsts showed a strong positive correlation with soil temperature ($p < 0.001$), with warmer conditions enhancing methanogenesis. In contrast, palsas showed no significant temperature correlation, likely due to moisture limitations restricting microbial activity. CO_2 fluxes exhibited a strong negative correlation with water level ($p < 0.001$) in thermokarsts, suggesting that periods of drought enhanced microbial decomposition and autotrophic respiration. This trend aligns with previous findings that low water levels promote oxygen diffusion, thereby accelerating aerobic decomposition. Interestingly, CO_2 fluxes did not correlate strongly with soil temperature, indicating that either hydrological constraints or other factors could have played a more dominant role in regulating respiration rates across sites. Collar design had a significant impact of fluxes, with deep-profile (D) collars exhibiting the strongest net CH_4 uptake (-4.40 mg CH_4 m^{-2} d^{-1}). This could suggest that oxygen penetration in deeper soils could have enhanced methane oxidation. For CO_2 fluxes, S collars exhibited the highest emissions (8.13 g CO_2 m^{-2} d^{-1}), followed by B collars (3.89 g CO_2 m^{-2} d^{-1}) and D collars (3.10 g CO_2 m^{-2} d^{-1}). Higher root respiration and microbial decomposition at the soil surface could explain higher fluxes, whereas lower CO_2 flux in D collars suggests that deeper soil layers may experience oxygen limitations, reducing aerobic microbial activity. Thermokarst development stages exhibited nonlinear CH_4 and CO_2 flux trends rather than a simple increasing or decreasing pattern. TM exhibited significantly higher CH_4 emissions (mean: 20.20 mg CH_4 m^{-2} d^{-1}) compared to TR and TY, suggesting that stable waterlogged conditions promoted methanogenesis. However, this contrasts with previous findings suggesting higher CH_4 fluxes in younger thermokarst bogs, highlighting the importance of site-specific hydrology.

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1 Introduction

1.1 Background

Permafrost has been described as ground that has remained at or below 0°C for at least two successive years (Biskaborn et al., 2019; Dobiński, 2020). These soils, however, are increasingly at risk of no longer conforming to their traditional definition. With increasing temperatures, it has been established that continuous permafrost zones in the Northern hemisphere have faced an increase in temperature of around 0.39°C ± 0.15 °C between 2007 and 2016 and rising (Biskaborn et al., 2019). This poses a threat to the stability of these perennially frozen grounds, altering their structure and making them susceptible to significant thawing over sustained periods of higher temperatures (Streletskiy et al., 2015). Permafrost thaw is characterized by increased temperature fluctuations in the soil, driven by the transition of ice to water. (E. A. G. Schuur et al., 2022). This can lead to progressive loss of ice content in the ground. Thawing of permafrost is of particular global importance due to one of the primary ecosystem functions that these soils provide, in the way of soil organic carbon (SOC) sequestration (Ping et al., 2015). Continuous (>90% area coverage) permafrost soils are mostly found in polar North Arctic circle, while discontinuous (50%-90%) and sporadic (10-50%) permafrost covering more of the sub-Arctic regions (Vincent et al., 2017).

A consequence of permafrost underlying a seasonally thawing layer would be that the soils are typically low temperature and with high moisture content, which is directly linked to the rate of decomposition of organic matter and other potential biogeochemical processes (Semenov et al., 2022; Y. Wang et al., 2020). This would result in lower decomposition rates and higher accumulation of organic matter, creating a rich storage of SOC which can date back to the Pleistocene (Knoblauch et al., 2021). The current estimate of coverage in the northern circumpolar permafrost region is at around 17.8×106 km^2 with the first 0-3 meters containing an approximate ~1000 Pg C (Hugelius et al., 2014). Although uncertainties remain for the estimate, it still contributes around 33% of the global pool of carbon (C) in only 15% of the global soil area, with potentially larger C contributions after the first 3 meters in depth (Hugelius et al., 2014).

Given these significant estimates of C susceptible to climate change and increasing temperatures, it is important to know the consequences on the structure and stability of permafrost soils. Upon extended periods of warming, thawing can result in the geomorphological transformation of the ground into thermokarst areas (Osterkamp et al., 2009). Thermokarst zones develop when the thawing of permafrost results in the degradation of the internal structure of the soil and subside unevenly within itself, leaving a water-logged depression of sediment in its stead (Kokelj & Jorgenson, 2013). A disruption in the thermal equilibrium of the permafrost, as a result of increasing temperature and snow cover, can alter ground characteristics and lead to an increase in frequency and magnitude of thermokarst terrains (B. M. Jones et al., 2011). The formation of such a landscape has drastic changes

in the hydrological regime of the area, given that thermokarst zones are a product of ice-rich ground which has melted and collapsed, it leaves anoxic depressions in the ground in the form of pits, gullies, and even lakes (Kokelj & Jorgenson, 2013).

Another consequence following the thawing of permafrost would be the deepening of the active layer thickness (ALT). The active layer (AL) would be a layer of the ground above the permafrost which is susceptible to annual variations of thawing and freezing periods (Liu et al., 2023). It is of particular importance as it is the layer in which most of the hydrological and biochemical processes occur and thus relevant with regards to the fate of stored C in the ground (Liu et al., 2023). The identification and measurement of the AL is dependent on temperature changes and thermal processes of the water phase (Dobiński, 2020). This can be related to the thickness of the layer on the ground which is subject to thawing during the summer period above permafrost. With Arctic regions warming up about two to three times faster compared to the global average, this has resulted in permafrost warming, and average ALT increases from 127.19 to 145.37 cm from 2000-2018, at a linear rate of 0.65 cm yr^{-1}, affecting around 71% of the total permafrost area (Bintanja, 2018; C. Li et al., 2022). Other studies have similarly observed mean ALT increases in the Northern Hemisphere at a linear rate of 0.86 cm yr^{-1} (Liu et al., 2023).

The repercussions of such drastic changes over time as a result of these different modes of thawing have left an uncertain future as to its widespread effect on the carbon cycle and the role of permafrost as a net source or sink of greenhouse gases (GHGs). CO_2 provides the largest carbon flux out of all the related greenhouse gases upon permafrost thaw; however, it is CH_4 that is estimated to lead to 40-70% of the total permafrost-related radioactive forcing in what remains of this century (Schneider von Deimling et al., 2015; Walter Anthony et al., 2018; Wittig et al., 2024). Even though CH_4 emissions are reportedly lower than that of CO_2, over a 100-year time scale, it can have a larger sustained-flux global warming potential around 45 times that of CO_2 (J. Li et al., 2024). These emissions of C from permafrost are controlled by several abiotic and biotic factors, including soil water content, air and ground temperature, ALT, and vegetation which will drive the microbial activity to enact correspondingly to its conditions (Masyagina & Menyailo, 2020; Wilson et al., 2017).

1.2 Importance and aim of study

1.2.1 Importance and aim of study

Permafrost contains twice as much C as the atmosphere, and thus its fate is crucial as to its future contributions and possible projections related to climate change (E. a. G. Schuur et al., 2015). Under business-as-usual warming scenarios, it is projected that emissions from CO_2 and CH_4 resulting from the thawing of permafrost in the Arctic, can contribute anywhere from 5-15% of the C stored in permafrost over the following decades and centuries (E. A. G. Schuur et al., 2022). This brings attention to the significance of expanding and filling gaps present within the current state of research

in permafrost peatlands. Peatland plateaus require specific environmental conditions for development, of which current suitable coverage is predicted to decrease by 76.3% under low-emission scenarios (RCP2.6) in the northern circumpolar region (Leppiniemi et al., 2023). While already in northern Norway, up to 70% of the area covered by peat plateaus and palsas has decreased since 1950 (Hichens-Bergström & Sannel, 2023). This past and current degradation of peatland and permafrost in the northern latitudes, alongside future projections, make the northern Fennoscandian region an area of interest to study permafrost peatlands and their response to thaw.

More specifically, observing and quantifying the gas fluxes of previously frozen C in northern Fennoscandian peatlands such as that in Finnmark, would expand the spatial coverage of currently existing research on GHG projections. This would be achieved with particular interest in the mode of thaw, which has been previously established to be an important variable affecting the vulnerability of SOC stocks (Biskaborn et al., 2019; Dobiński, 2020; Estop-Aragonés, Czimczik, et al., 2018; B. M. Jones et al., 2011).

Therefore, the aim of this study is to quantify the response of permafrost peatland in Finnmark, Norway to different modes of thawing (thermokarst & active layer deepening) with specific regard to the fluxes of CO_2 and CH_4. Under these modes of thaw, various biotic and abiotic factors will be considered, which characterize areas dominated with thermokarst and those with notable active layer deepening. In order to achieve this purpose, an in-situ closed chamber method was followed in order to measure gas fluxes from the permafrost peatland sites. This study will build upon research done in other study sites in Canada, Alaska, and Siberia, and add valuable insights into the fate of previously frozen SOC to thawing (Fewster et al., 2022; Heffernan et al., 2020; M. C. Jones et al., 2017; O'Donnell et al., 2012).

1.3 State of the art

1.3.1 Effect of soil moisture on SOC

One of the limitations of microbial activity with regards to the production and release of GHGs is that of soil moisture (Shah et al., 2024). High moisture contents in the soil leads to an increasingly anaerobic environment, thus promoting the activity of such specialized microbial communities to decompose organic matter and produce CH_4 in through methanogenesis (Shah et al., 2024). Methanogenic bacteria can utilize substrates such as dead plants and roots and already existing dissolved and particulate organic carbon in order to produce CH_4 as a product of organic matter decomposition (Shah et al., 2024). This form of heterotrophic respiration especially thrives under limitations in electron acceptors like sulfate and nitrate present in the soil, which favors methanogens with freely available labile carbon (X. Song et al., 2020; Sutton-Grier et al., 2011).

Under lower soil moisture conditions, or in increasingly accessible aerobic conditions nearer to the soil surface, an antithetical reaction is present in the form of CH_4 consumption by methanotrophs (Lee

et al., 2012). CH_4 is consumed near the surface upon its transport pathway upwards from deeper in the soil. If the water table is low in a peatland, it leaves the soil exposed to aerobic conditions which disrupt the general pathway for the transport of CH_4, which would be through molecular diffusion upwards into the overlying water column (L. Wang et al., 2021). Anywhere from 20-60% of CH_4 produced can be oxidized by methanotrophic communities in just a few centimeters worth of oxic zones in an overlaying water column in the soil (Miner et al., 2022). Other forms of CH_4 release upon production come in the form of vascular plant-mediated transport and ebullition, both of which are also influenced by soil moisture and position of the water table (Klapstein et al., 2014; Olefeldt et al., 2013; L. Wang et al., 2021). Through vascular plants, certain tissues, such as that of aerenchyma, are able to create a direct pathway for CH_4 release without being heavily affected by methanotrophic bacteria in aerobic zones (Knox et al., 2021; Wilson et al., 2017). Such vascular plants can have a two-sided effect, with them also allowing oxygen to be transported from the surface to the roots and thus creating aerated zones deeper in the soil column (Knox et al., 2021). Finally, ebullition is also a process of significance as it is the formation and release of accumulated gas bubbles deeper in sediments under anaerobic conditions (Klapstein et al., 2014; L. Wang et al., 2021). This form of CH_4 release might not be the most frequent and general of pathways, however, it does represent a large percentage of the overall emissions of CH_4 (L. Wang et al., 2021). Previous studies and reviews have suggested that ebullition can represent anywhere from 30-95% of total CH_4 emissions, with high variability due to the episodic nature of this process, posing challenges in accurately establishing the temporal trends (Chingangbam & Khoiyangbam, 2024; Liikanen et al., 2003; Miller et al., 2017; L. Wang et al., 2021; L. Zhang et al., 2020). Given its potentially large percent contribution, it makes it an important process for heavily saturated soils such as those found in thermokarst lakes and wetlands, with them accounting for approximately 25–35 % of global methane budget (Z. Zhang et al., 2025).

Thawing of permafrost, creating increasingly aerated environments may inhibit or limit CH_4 production, but it does promote the release of CO_2 (Schädel et al., 2016). Emissions from this GHGs have been found to be highest under a moisture content of 70%, with lower percentages also proving favorable for soil respiration to occur (Rodenhizer et al., 2023). Aerobic conditions lead to the oxidation of CH_4 and in consequence, to the production of CO_2 from these zones. This is one of the contributing pathways and sources of CO_2 in peatlands, yet altogether, under incubation experiments, it can be shown that CO_2 can produce up to 6 orders of magnitude more emissions in contribution to total C compared to CH_4 (Bond-Lamberty et al., 2016).

As the active layer in permafrost thickens, moisture is allowed to infiltrate deeper into the soil column and thus result in the drying of the near surface zones (Andresen et al., 2020). This would result in increased emissions of CO_2 from the ground due to root respiration and organic matter decomposition (Lee et al., 2010). Upon significant permafrost thaw however, the contribution of microbial decomposition is more pronounced with more readily available C. A 13-year warming experiment

which observed CO_2 fluxes following varying rates of subsidence, showcased how in slower subsiding plots, greater CO_2 fluxes were produced (Rodenhizer et al., 2023). While in more abrupt subsidence scenarios, whereby the plot was able to get waterlogged at a faster rate, forming thermokarst features, exhibited lower carbon release (Rodenhizer et al., 2023). Following thawing processes, a more exposed soil column may release previously trapped CO_2 found deeper in air pockets, which can then lead to an increase in emissions early on in the growing season. Other studies also support the depth dependence of soil moisture and its influence on gas fluxes, with CO_2 emissions being significantly higher in the first 0-10 cm compared to deeper in the soil (Lee et al., 2010). Even though CO_2 was found in higher concentrations deeper in the soil column, the low diffusivity in soil with higher soil moisture limited CO_2 flux by 2 orders of magnitude (Lee et al., 2010).

1.3.2 Effect of temperature on SOC

Permafrost carbon feedback is the positive feedback loop formed when a rise in temperature leads to thawing of permafrost and organic matter decomposition, thereby releasing GHGs into the atmosphere and thus exacerbating anthropogenic warming (Schneider von Deimling et al., 2012). This rise in temperature thus becomes one of the key drivers influencing the fate of C stored within a soil column, and fluctuations from such a variable can affect CO_2 and CH_4 differently (Bond-Lamberty et al., 2016; Y. Song et al., 2021; X. Wang et al., 2010).

In general, increasing temperatures can drive bacterial and fungal activity in the soil, and therefore also increase available nutrients and support further emissions of CO_2 (Felice et al., 2024; Seaton et al., 2022). Lab incubation experiments, manipulating different temperatures have shown that increased warming in the soil increases the availability of the soil carbon and nitrogen (Lu et al., 2022). Moreover, the results showed that an increase in soil temperature by 10 °C, also resulted in elevated rates of emissions for CO_2 by 5.1-9.4 times and 3.8-6.4 times for CH_4 (Lu et al., 2022). These observations were attributed to the stimulation of microbial activity within the soil with readily available nutrients which thus fueled higher rates of soil carbon emissions. Other studies have discovered similar results in the effect of increasing temperature, with CO_2 emissions being 3.36 and 2.74 times larger in top and subsoils, respectively at 15 °C compared to soils at 5 °C (Y. Song et al., 2021). However, it is CH_4 that is shown to be more responsive to changes in temperature, with emissions being 34 and 83 times greater in top and subsoils, respectively in non-waterlogged conditions of the soil (Y. Song et al., 2021). This sensitivity from CH_4 to temperature would be as a result of increased methanogenic community size and activity at higher temperatures. Furthermore, general microbial activity and microbial respiration entropy are also increased with temperature which can lead to higher CO_2 production (Lu et al., 2022).

Different modes of thaw can also have different properties with regards to temperature. Some studies have shown that permafrost, which has thawed into a thermokarst and developed subsided zones,

have also presented higher soil temperatures compared to less subsided zones (Natali et al., 2015). This can be the case especially during winter months when snow cover can insulate the soil among the previously loped depression. This can then have a cascading effect on soil temperatures into the summer.

The deepening of the active layer following the thawing of permafrost from higher temperatures can also have compounding effects on the production of CO_2 in the soil. This can come in the form of not only microbial decomposition of organic matter but also root respiration (Jin & Ma, 2021). This deepening of the active layer, enhanced by the permafrost carbon feedback can lead to scenarios where Arctic permafrost can switch from a current net sink of CO_2 with 0.3–0.6 Pg C per year to a net source by the end of the decade (Jin & Ma, 2021).

1.3.3 Effect of vegetation on SOC

Heterotrophic and autotrophic respiration can yield different results with regards to CO_2 and CH_4 emissions, and so the influence of vegetation cannot go unnoticed. Plant communities have been identified to be key drivers in CH_4 fluxes, mainly as a result of the role that vascular plants have in the transport mechanisms of CH_4 produced by methanogenic bacteria deeper in the soil column (Knoblauch et al., 2018, 2021). The presence of sedges in the soil can allow the presence of root structures supplying vascular transport for CH_4 and thus limiting oxidation by methanotrophs as it makes its way into the atmosphere (Klapstein et al., 2014; E. a. G. Schuur et al., 2015).

Other studies have even highlighted vegetation cover, more specifically sedges, as the primary controlling factor in terms of CH_4 fluxes in permafrost areas (Olefeldt et al., 2013). Under similar environmental conditions, sites that had a greater sedge cover presented higher CH_4 emissions, with aerenchymatous roots and stems being of great importance in the processes involving CH_4 production and transport (Olefeldt et al., 2013). Moreover, CH_4 production was also enhanced with root exudation being a source of organic acids in the soil which also aided in the production of CH_4 (Waldo et al., 2019). Thus, vegetation as a whole can have pronounced effects, particularly in CH_4 fluxes, with other studies highlighting a 22-66% associated increase from 1970-2000 originating from a mire of interest (Christensen et al., 2004).

The growth and establishment of diverse plant communities following permafrost thaw can alter soil temperature and moisture dynamics, sometimes counteracting previously observed fluctuations. Vegetative colonization in thermokarst depressions and zones with thicker active layers may facilitate the uptake of CO_2 from the atmosphere, continuing carbon sequestration processes (E. a. G. Schuur et al., 2015). While the overall impact of these changes on the carbon cycle remains uncertain, new plant growth driven by longer growing seasons and increased air and soil temperatures could enhance primary productivity and atmospheric carbon uptake (E. a. G. Schuur et al., 2015).

1.4 Hypotheses

Following the consideration of the main influential factors that determine the fate of SOC in thawing permafrost, several hypotheses can be established as to match the aim of the study:

1) CO_2 flux will be comparatively higher in palsas than in thermokarst features, whereby it will exhibit lower soil moisture content with its progressive active layer deepening. This creates aerobic conditions which expose large stocks of previously frozen organic matter to decomposition. Contrastingly, CH_4 flux will be higher in thermokarst features compared to palsas, due to the waterlogged, anoxic nature of its formation. This presents anaerobic conditions favorable for methanogenic bacteria and transport mechanisms like ebullition.

2) CO_2 and CH_4 fluxes will be higher in soils with higher soil temperature conditions. Higher temperatures enhance microbial activity, accelerating the decomposition of organic matter. This process increases nutrient availability, stimulating microbial respiration and methanogenesis, which in turn elevate greenhouse gas emissions.

3) CO_2 and CH_4 fluxes will vary between palsa collar treatments, with higher fluxes in deep collars compared to sealed bottom collars due to greater soil profile exposure and contributions from previously frozen SOC. Vegetated surface collars will show distinct fluxes due to plant-mediated transport and contributions from root respiration and exudates.

4) CO_2 and CH_4 fluxes will differ among thermokarst development stages (transitionary, recent, young, mature) due to variations in soil hydrology, vegetation colonization, and organic matter composition. Advanced stages (mature thermokarst) are expected to show higher CO_2 and CH_4 emissions due to established methanogenic microbial communities and greater cumulative plant respiration.

2 Materials and methods

2.1 Site description

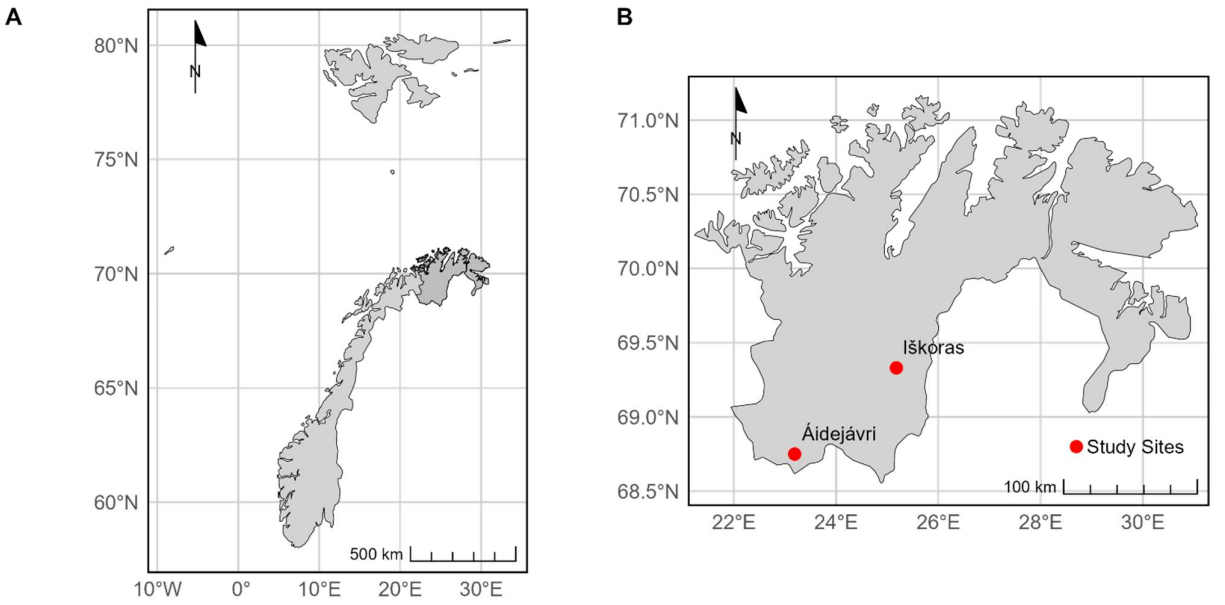


Figure 1. *(A) Map of Norway, with the Finnmark county highlighted in dark grey. (B) Close up map of the Finnmark county, showing the locations selected for measurements that display disturbance by thermokarst formation and active layer deepening (Iškoras: 69°20' N 25°18' E and Áidejávri: 68°45' N 23°19' E).*

The selected sites were located within the county of Finnmark (Norway) including Iškoras (69°20' N 25°18' E) and Áidejávri (68°45' N 23°19' E) as seen from Fig. 1. Here, the dominating features are mires, which give way to peat plateaus and palsas thus representing a sporadic distribution of permafrost within this region (Gisnås et al., 2017; T. Zhang et al., 1999). Both sites are composed of minerotrophic peatland fens which have been formed through the process of paludification (Sannel et al., 2018), showing different locations with palsas and adjacent thermokarst features from which the experimental design was implemented on. The Iškoras site is located at 360 m a.s.l and is characterized by the thermokarst dominated features and surrounding glacial fluvial deposits. It is an already established site with multiple active research bodies with continuous monitoring activities (NTNU, University of Oslo, NIVA-Norwegian Water Research Institute, and NORCE-Norwegian Research Centre). Áidejávri is placed at 370 m a.s.l. and is characterized by the extensive peatland plateau and varying degrees of active layer deepening several thermokarst formations with variable ground subsidence. It is a recently explored site, identified by Sebastian Westermann of the University of Oslo.

Both sites share a similar spread of vegetation, primarily consisting of low shrubs like *Betula nana* (Dwarf birch), *Ledum palustre* (Wild rosemary), *Empetrum nigrum* (Crowberry), *Vaccinium vitis-idaea*

(Lingonberry), *Andromeda polifolia* (Bog rosemary), and *Rubus chamaemorus* (Cloudberry). These plants reflect the nutrient-poor and acidic soil conditions which are typical of permafrost peatlands (Jonasson, 1981; Shaver, 1983). Variation within these sites from palsa to thermokarst is also visible from the transition in vegetation as the collapsed palsas give way to waterlogged depressions. Transitional vegetation such as *Vaccinium uliginosum* (Bog bilberry) was also found as wetter conditions took over and the thermokarst features started appearing, with sedge and moss dominated vegetation like *Carex aquatilis* (Water sedge), *Sphagnum fuscum* (Rusty bog moss), *Eriophorum vaginatum* (Tussock cottongrass), *Sphagnum lindbergii* (Lindberg's bog moss) and possibly *Mylia anomala* (Anomalous flapwort).

2.2 Experimental design

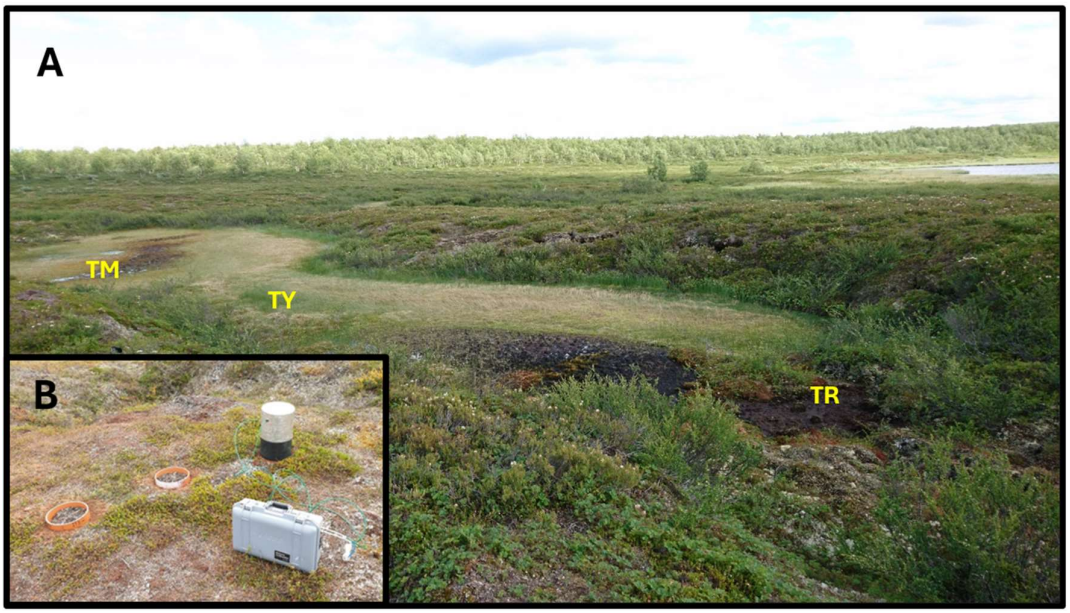


Figure 2. *(A) Áidejávri thermokarst feature with differing stages of development including Recent Thermokarst (TR), Young Thermokarst (TY), and Mature Thermokarst (TM). (B) Gas flux measurement in a palsa in Iškoras, showing the B, D and S collars (chamber on S).*

After an exploration of the physical landscape of both sites, several spots were identified and selected for the trenching and installation of the collars to be used for the measurements. The locations selected were based on preliminary thaw depth measurements along the palsas using a probe rod in order to see the current depth of the thawing at the date of arrival and the feasibility of the collar installation. Final locations for the palsas were chosen based on the criteria that: a) Thaw depth was greater than 30 cm. b) Didn't show extensive signs of degradation or near subsidence in the form of cracks on the ground or too close proximity to a thermokarst. c) Displayed variations in vegetation within close proximity, allowing for the implementation of different treatments.

Thermokarst locations were chosen with the basis of a potential chronosequential gradient, representing the stages of permafrost thaw and thermokarst development. These were identified and selected through the gradual change in vegetation presence and color (Estop-Aragonés, Czimczik, et al., 2018). This gradient was established as the recent (TR), young (TY), and mature (TM) stages of a thermokarst (Figure 2A). TR was characterized as completely unvegetated thick peat material under complete waterlogged conditions. TY was characterized as a highly water saturated section of the thermokarst with a high density of sphagnum-dominated communities in the form of a floating mat. TM was characterized as a late-stage development of the thermokarst with now more densely decomposed material across a depth gradient with a darker color tone. In Iškoras, an earlier stage of thermokarst development was selected as it showed to have a different type of vegetation compared to the other thermokarst and palsas. Thus, this transitionary stage was dominated more by species of lichens from the genus *Cladonia* and a potential indicator of an incomplete hydrological transition (Cazaurang et al., 2023).

Once potential locations were selected and met the criteria, 47 collars made of PVC (internal diameter 23.5 cm) were inserted across 20 different locations. From these, 32 collars were in Áidejávri, divided as follows: 12 Palsa (P), 4 TR, 3 TY, and 3 TM. In Iškoras, the 15 collars consisted of: 9 P and 6 early thermokarst (T). A further division of the collars was implemented, whereby there were 20 sealed-bottom (B), 20 full profile deep (D), and 7 surface (S) collars. This division was done to establish different treatments that can affect the gas flux measurements. B, D, and S collars shared the same internal diameter, however, differed in height, as B and D collars had a height of 35 cm and were inserted 30 cm into the soil, while S collars were 15 cm in height and were inserted around 10 cm into the soil. This collar insertion depth was chosen as the active layer depth was predicted to be around 50 – 100 cm and so this depth would encapsulate the fluxes from this section (Estop-Aragonés, Cooper, et al., 2018; Estop-Aragonés, Czimczik, et al., 2018). The B collars were sealed at the bottom in order to exclude potential C sources from beneath the depth of interest. Small holes were also drilled in order to aid drainage in the B collars. The insertion of the collars was done by cutting into the soil using a serrated knife, extracting a peat monolith which would fit the specifications of the B collar and then reinserted into the ground. B and D collars were inserted in an unvegetated section of the palsa, while the S collars were inserted into nearby vegetated sections. These vegetated sections included the scattered low shrub species previously mentioned. In the thermokarst, only B and D collars were inserted, all within accessible distance from wooden boards, and all newly grown green shoots were continuously removed throughout the growing season (Estop-Aragonés, Cooper, et al., 2018).

Environmental and soil physical conditions were measured using hand-held probes for soil temperature, soil moisture (ThetaProbe ML2x) at 0-6 cm measurement depth, and atmospheric pressure. These measurements were conducted following every gas flux measurement.

2.3 CO_2 and CH_4 flux determination

CO_2 and CH_4 concentrations were determined through the use of portable greenhouse gas analyzers LI-7810 (LI-COR Biosciences, Lincoln, NE, USA) provided by Prof. Dr. Hanna Lee from NTNU and the Ultra-Portable Greenhouse Gas Analyzer (UGGA) (Los Gatos Research, San Jose, CA, USA) provided by Uni Münster. Measurements were made by sealing the collars with a dark PVC chamber (15 L) and a rubber tube for 5 minutes or longer, during which the concentrations were recorded at 1 s intervals (Figure 2B). Gas flux measurements started approximately two weeks after the installation of the collars, aiming to minimize any potential disturbances (Gibson et al., 2019). These measurements were conducted weekly throughout the growing season, from early July until early September. In total, 9 measurements were done per site for all 47 collars. In order to determine the fluxes of CO_2 and CH_4, a linear regression of the increase or decrease in these gas concentrations over time was done to be able to attain a slope with an R^2 value of > 0.90 to be used as flux rates. The gas flux F_{gas} for CO_2 and CH_4 was calculated in units of mmol m^{-2} d^{-1} using the following equation (Muhr et al., 2008):

$$F_{gas} = \left(\frac{dc}{ct}\right) \cdot \left(\frac{M_C \cdot V_H \cdot 1440\ min\ d^{-1}}{M_V \cdot A_H \cdot 1000\ ppm}\right) \cdot \left(\frac{P_a}{P_N \cdot (1 + 0.00366 \cdot T_a)}\right) \quad (1)$$

Here, $\left(\frac{dc}{ct}\right)$ represents the change in gas concentration over time, measured in ppm min^{-1}, within the mesocosm headspace. M_C is the molar mass of carbon for the measured gas (e.g., 12 g mol^{-1} for CH_4, 44 g mol^{-1} for CO_2), and V_H is the volume of the mesocosm chamber in liters. The flux is normalized to the soil surface area $A_H$within the chamber, expressed in m^2, and adjusted for atmospheric conditions using the measured air pressure P_a and the standard air pressure (P_N = 1013 hPa). Temperature adjustments are applied using the factor $(1 + 0.00366 \cdot T_a)$, where T_a is the measured air temperature in °C, based on the relationship between absolute and Celsius temperature scales. The constant 1440 accounts for the conversion from flux per minute to flux per day (60 $minh^{-1}$ × 24 $hday^{-1}$), and the factor 1000 converts concentrations from mg to mmol. For graphical presentation, the units used were g CO_2 m^{-2} d^{-1} and mg CH_4 m^{-2} d^{-1}.

2.4 Statistical analysis

All statistical analysis was carried out in R (Version 4.4.1) (R Core Team, 2024). A number of R packages were employed to ensure robust handling, modeling, and visualization of the data acquired from the field. Initially, the readxl (Hadley Wickham & Jennifer Bryan, 2023) and dplyr (Hadley Wickham et al., 2023) packages were used in order to import and clean the data into an appropriate structure for later analysis. This facilitated the process of identifying and handling possible missing values, non-applicable data, the distribution of the data and potential outliers. The greenhouse gas flux data (CO_2 and CH_4) were analyzed using linear mixed-effects models (LMMs) from the glmmTMB

(Mollie E. Brooks et al., 2017) and lme4 (Douglas Bates et al., 2015) packages. These models aided in accounting for the hierarchical structure of the dataset by incorporating both fixed and random effects. Fixed effects included factors such as location, treatment, and site, while random effects captured variability among the individual collars and the measurement dates. Model diagnostics were conducted using the DHARMa (Florian Hartig, 2024) package to evaluate residual normality, homoscedasticity, and independence, ensuring the validity of the results. Due to the variable nature of gas flux measurements, the data for both CO_2 and CH_4 were not normally distributed, and so the Shapiro-Wilk test was used in order to determine which transformation method effectively normalized the data distributions. For CO_2, the logarithmic and square root transformations were insufficient to normalize the CO_2 flux data, as both exhibited significant deviations from normality ($W = 0.954$, $p < 0.001$ and $W = 0.967$, $p = 0.002$, respectively). A Box-Cox transformation, from the car (John Fox & Sanford Weisberg, 2019) package, resulted in no significant deviation from normality and thus was used for proceeding statistical analysis. For CH_4, the outcome was similar, with logarithmic and square root transformations showing significant deviations ($W = 0.559$, $p < 0.001$ and $W = 0.500$, $p < 0.001$, respectively), while the Box-Cox transformation partially improved the normality, but still showed deviation ($W = 0.888$, $p < 0.001$). The Box-Cox transformation, as described by Box and Cox (1964), was applied to the CO_2 and CH_4 flux data to address issues of non-normality and heteroscedasticity. This method estimates an optimal transformation parameter (λ) that minimizes skewness and improves data symmetry. Therefore, using the powerTransform function in R, the optimal λ was calculated separately for CO_2 and CH_4 flux data. For CH_4, an offset of +1 was added to ensure all values were positive, as required by the Box-Cox transformation. The transformed values were then computed using the bcPower function, with the resulting variables exhibiting distributions closer to normality. Nevertheless, due to the highly non-normal distribution of CH_4 data, outliers in the CH_4 flux data were removed using the interquartile range (IQR) method. The IQR, representing the range between the first quartile (Q1) and third quartile (Q3), was used to define boundaries for acceptable values:

$$Lower\ Bound = Q1 - (1.5\ x\ IQR),\ Upper\ Bound = Q3 + (1.5\ X\ IQR) \quad (2)$$

Data points outside these bounds were excluded to reduce the influence of extreme values and improve the reliability of the analysis. This step was necessary to address persistent non-normality in the CH_4 flux data and ensure that the data has satisfied the criteria for the LMMs.

Post-hoc analyses were conducted using the emmeans (Russell V. Lenth, 2024) package to evaluate pairwise comparisons which were deemed important to address the hypotheses. These analyses allowed for a more detailed interpretation of within group-level differences and their implications for greenhouse gas flux dynamics. To investigate the relationships between multiple variables and

identify key drivers of greenhouse gas fluxes, a Principal Component Analysis (PCA) was performed. The PCA was conducted using the FactoMineR (Sébastien Lê et al., 2018) package, and the results were visualized with factoextra (Alboukadel Kassambara & Fabian Mundt, 2020). Additionally, the ggrepel (Kamil Slowikowski, 2024) package was used to ensure clear and effective labeling of data points in PCA plots. All plots were created using the ggplot2 (Hadley Wickham, 2016) package, with additional layout customization and plot merging achieved through cowplot (Claus O. Wilke, 2024), gridExtra (Baptiste Auguie, 2017), and ggspatial (Dewey Dunnington, 2023). Maps of the study sites (Fig.1) were developed using the sf (Edzer Pebesma, 2018), rnaturalearth (Philippe Massicotte & Andy South, 2023), and rnaturalearthdata (Andy South et al., 2024) packages to provide geographic context. The broom.mixed (Ben Bolker & David Robinson, 2024) package was used to organize the model outputs, facilitating their integration into tables.

3 Results

3.1 Temporal Patterns of Environmental and Gas Flux Variables

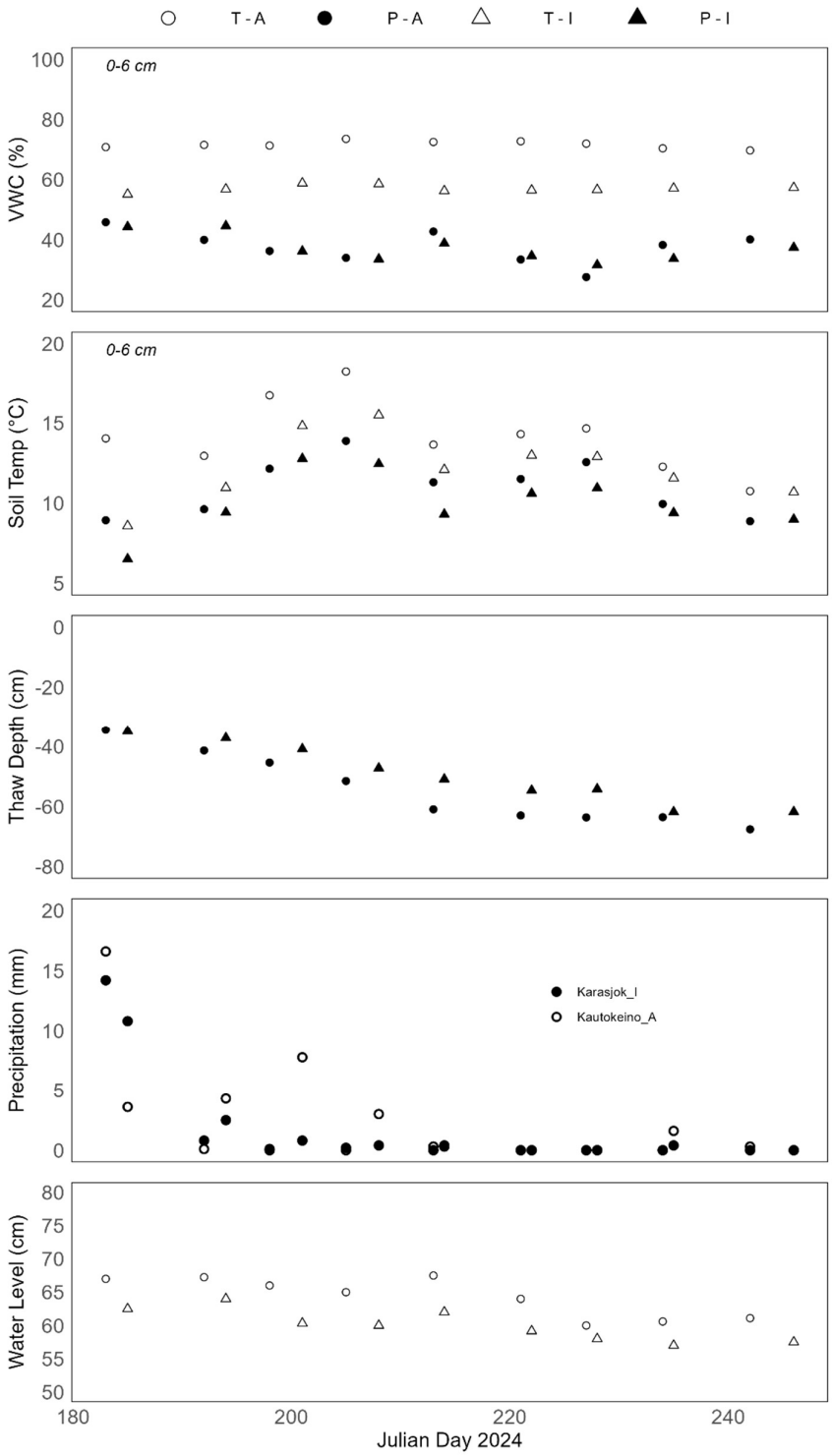


Figure 3. *Temporal variations in mean VWC, Soil Temperature, Thaw Depth, Precipitation, and Water Level across locations and sites during the 2024 growing Season. The legend indicates thermokarst (T), palsa (P), Site A (A), and Site I (I). Karasjok and Kautokeino represent closest weather stations to Site I and A, respectively.*

Environmental variables exhibited distinct temporal patterns across palsa and thermokarst locations with significant differences between Site A and Site I during the 2024 growing period spanning from July 1st to September 2nd, 2024.

Volumetric water content (VWC) was higher in thermokarst sites (68.2 ± 6.5%) compared to palsa sites (37.4 ± 11.5%). Within thermokarst sites, Site A exhibited significantly higher VWC (71.6%) compared to Site I (65.7%, $p < 0.0001$). In contrast, palsa sites showed no significant difference in VWC between Site A (37.6%) and Site I (37.2%, $p = 0.82$). Time-series analysis indicated no significant seasonal trend in VWC for thermokarst sites ($p = 0.26$), whereas palsa sites exhibited a weak but significant decrease in VWC over time ($p = 0.0043$).

Soil temperature varied over the growing season, with thermokarst sites exhibiting higher mean temperatures (13.7 ± 2.3°C) than palsa sites (10.6 ± 2.0°C). Within thermokarst locations, Site A (TR, TY, TM) had a significantly higher mean soil temperature (14.2 ± 2.2°C) than Site I (T) (12.2 ± 2.3°C, $p < 0.0001$). In palsa locations, Site A exhibited a higher mean soil temperature (11.0 ± 2.0°C) compared to Site I (10.2 ± 1.8°C, $p < 0.0001$). Despite both thermokarst and palsa sites following a similar seasonal pattern, time-series analysis detected a significant seasonal increase in soil temperature in thermokarst sites ($p < 0.0001$), while no statistically significant trend was observed in palsa sites ($p = 0.79$).

Thaw depth, measured exclusively in palsa sites, varied throughout the growing season, with a mean depth of 54.6 ± 12.0 cm at Site A and 49.3 ± 10.2 cm at Site I. Thaw depth was significantly greater in Site A compared to Site I ($p = 0.0012$). The maximum recorded thaw depth reached 68.2 cm. Time-series analysis confirmed a strong seasonal increase in thaw depth ($p < 0.0001$).

Water levels, recorded only in thermokarst sites, fluctuated throughout the season with a mean of 0.62 ± 0.03 m. A significant difference was observed between Site A and Site I ($p < 0.0001$), where Site A had lower water levels (0.60 ± 0.02 m) compared to Site I (0.63 ± 0.03 m). However, despite a general decline, no significant seasonal trend was detected in water level variation ($p = 0.26$).

Precipitation varied over the growing season, with an initial peak on July 1, 2024, when daily precipitation reached 16.6 mm at Site A and 14.2 mm at Site I. Following this event, precipitation levels declined, with intermittent rainfall occurring throughout the season. The mean daily precipitation was 1.99 mm at Site A and 1.70 mm at Site I. Smaller precipitation peaks were recorded on July 10, July 18, and August 23, though they did not exceed the early-season maximum.

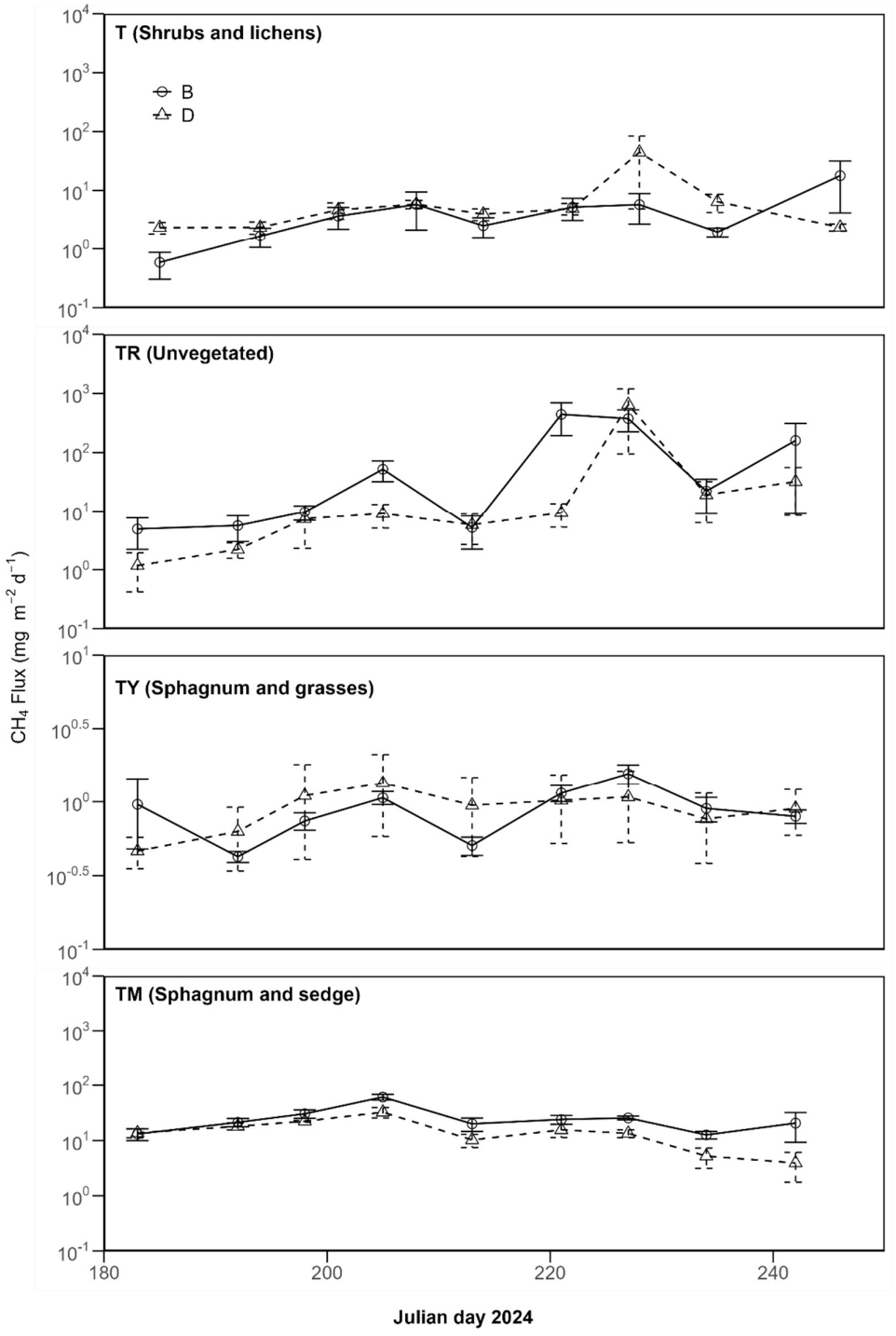


Figure 4. *Temporal variations in CH_4 flux across thermokarst development stages (T, TM, TR, TY) during the 2024 growing season. Error bars indicate standard error of the mean. Collar treatment presented as bottom (B) and deep (D). Log scale was utilized for improved visualization.*

Figures 4 and 5 illustrate seasonal trends in CH_4 and CO_2 fluxes across thermokarst development stages and collar treatments over time. CH_4 fluxes displayed high variability, with all stages peaking mid-season. TR recorded the highest CH_4 flux (100.51 mg CH_4 m^{-2} d^{-1}), followed by TM (20.20 mg CH_4 m^{-2} d^{-1}), T (6.82 mg CH_4 m^{-2} d^{-1}), and TY (0.91 mg CH_4 m^{-2} d^{-1}). Collar treatment effects on CH_4 flux were variable across sites. T recorded higher CH_4 fluxes in D collars, with a difference of +3.62 mg CH_4 m^{-2} d^{-1} compared to B collars. In contrast, TM and TR showed higher CH_4 emissions in B collars, with differences of +10.43 mg CH_4 m^{-2} d^{-1} and +38.12 mg CH_4 m^{-2} d^{-1}, respectively. TY exhibited negligible differences between collar treatments (+0.02 mg CH_4 m^{-2} d^{-1}).

CO_2 fluxes followed a more gradual seasonal increase, generally peaking between mid and late July across all stages. TM exhibited the highest mean CO_2 flux (5.72 g CO_2 m^{-2} d^{-1}), followed by T (5.16 g CO_2 m^{-2} d^{-1}), TY (5.02 g CO_2 m^{-2} d^{-1}), and TR (3.55 g CO_2 m^{-2} d^{-1}). Collar effects on CO_2 flux followed a different trend from CH_4. Across most thermokarst sites, B collars recorded higher CO_2 fluxes than D collars. In TR, B collars averaged 4.04 g CO_2 m^{-2} d^{-1}, compared to 3.06 g CO_2 m^{-2} d^{-1} in D collars (+0.99 g CO_2 m^{-2} d^{-1} difference). Similar patterns were observed in TM and TY, where B collars recorded 0.60 g CO_2 m^{-2} d^{-1} and 2.78 g CO_2 m^{-2} d^{-1} higher emissions than D collars, respectively.

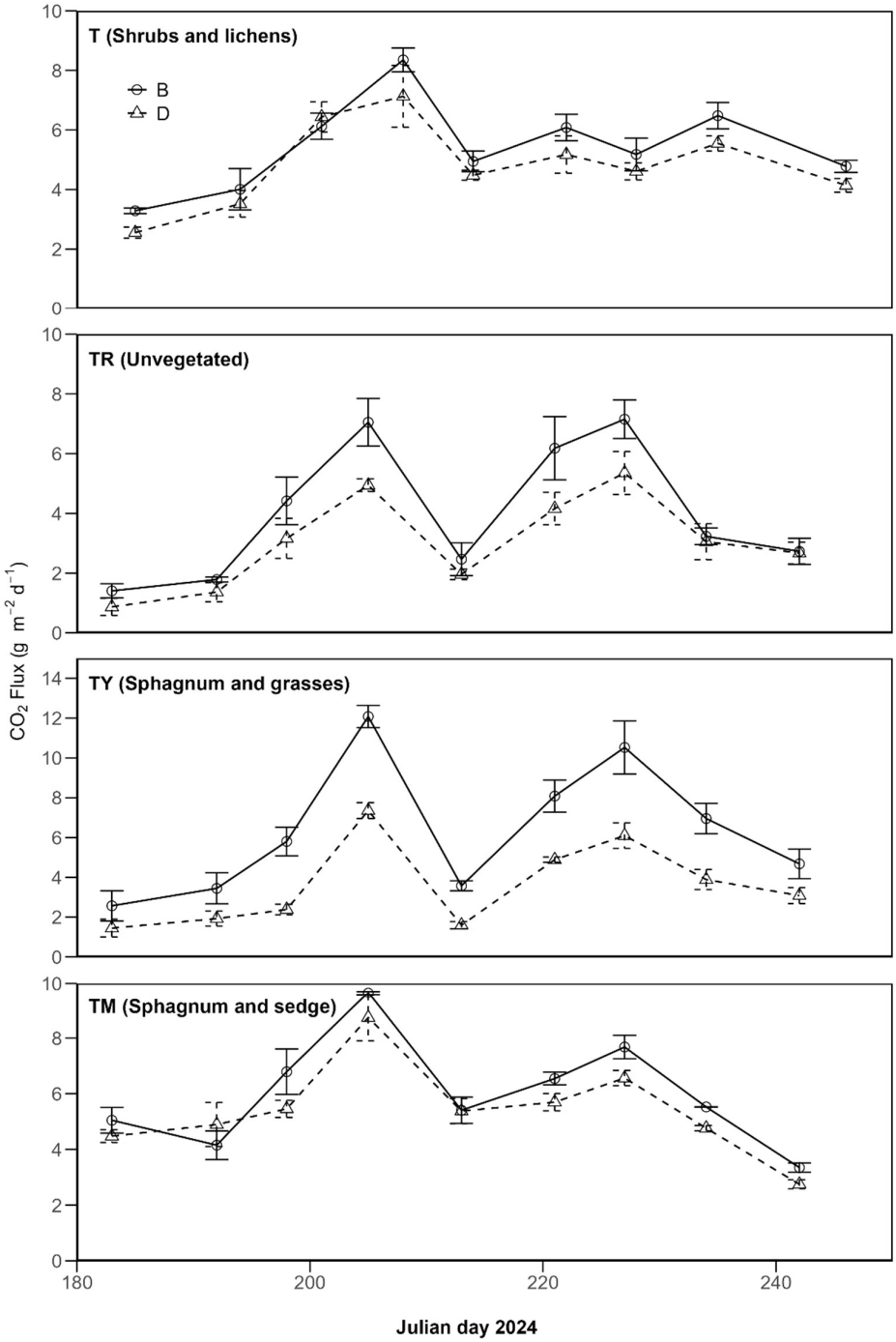


***Figure 5**. Temporal variations in CO_2 flux across thermokarst development stages (T, TM, TR, TY) during the 2024 growing season. Error bars indicate standard error of the mean. Collar treatment presented as bottom (B) and deep (D).*

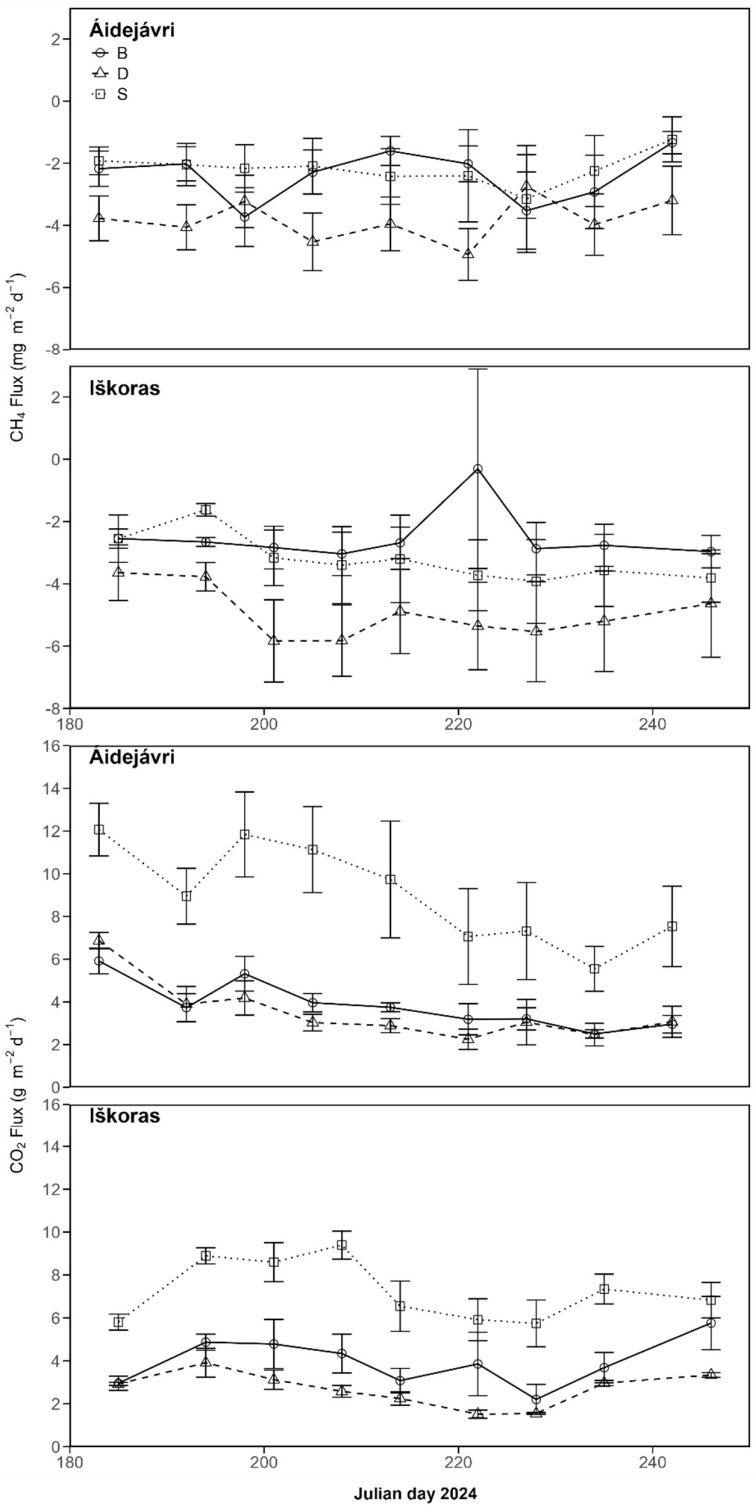


Figure 6. *Temporal variations in CH_4 flux and CO_2 flux across Sites A and I in palsas during the 2024 growing season. Error bars represent standard error of the mean. Collar treatments are indicated as bottom (B), deep (D) and surface (S). Panels distinguish between sites to highlight spatial variability.*

Figure 6 highlights the temporal variations in CH_4 and CO_2 fluxes across Áidejávrri (Site A) and Iškoras (Site I) in palsas during the measurement period, with panels distinguishing spatial variability between sites. At Site A, CH_4 fluxes were highest in S collars (-2.19 mg CH_4 m^{-2} d^{-1}), followed by B collars (-2.40 mg CH_4 m^{-2} d^{-1}) and D collars (-3.82 mg CH_4 m^{-2} d^{-1}). At Site I, CH_4 uptake was slightly stronger overall. D collars recorded the most negative flux (-4.97 mg CH_4 m^{-2} d^{-1}), followed by S collars (-3.22 mg CH_4 m^{-2} d^{-1}) and B collars (-2.52 mg CH_4 m^{-2} d^{-1}). CO_2 fluxes exhibited positive emissions across all sites and treatments, with notable differences between collars. At Site A, S collars recorded the highest fluxes, averaging 9.02 g CO_2 m^{-2} d^{-1}, while B collars and D collars showed lower fluxes of 3.83 g CO_2 m^{-2} d^{-1} and 3.52 g CO_2 m^{-2} d^{-1}, respectively. At Site I, CO_2 fluxes were slightly lower overall but followed a similar trend. S collars recorded the highest emissions at 7.23 g CO_2 m^{-2} d^{-1}, followed by B collars at 3.95 g CO_2 m^{-2} d^{-1}, and D collars at 2.68 g CO_2 m^{-2} d^{-1}.

3.2 Greenhouse gas fluxes across palsa and thermokarst features

Table 3-1. *Linear mixed-effects model summaries for Box-Cox-transformed CH_4 (left) and CO_2 (right) fluxes comparing palsas and thermokarst features during the 2024 growing season. Fixed effects include Location (T, TM, TR, TY; intercept = P) and collar Treatment (D; intercept = B). Estimates represent the direction and magnitude of the effect on gas flux.*

Predictor	Estimate	Std. Error	p-value
CH4			
(Intercept)	0.23	0.01	0.00***
LocationT	0.16	0.02	0.00***
LocationTM	0.27	0.02	0.00***
LocationTR	0.22	0.02	0.00***
LocationTY	0.11	0.02	0.00***
TreatmentD	-0.03	0.01	0.00**
TreatmentS	0.01	0.02	0.58
CO2			
(Intercept)	9.08	0.33	0.00***
LocationT	1.76	0.32	0.00***
LocationTM	1.83	0.30	0.00***
LocationTR	-0.36	0.34	0.29
LocationTY	0.95	0.38	0.01*
TreatmentD	-0.82	0.20	0.00***
TreatmentS	3.07	0.35	0.00***

Note:

Significance levels: *** p < 0.001, ** p < 0.01, * p < 0.05. Estimates are derived from Box-Cox transformed flux values.

The linear mixed-effects model results for CH_4 flux (Table 1) indicated significantly higher emissions across all thermokarst stages (T, TM, TR, TY) compared to palsas, with estimates ranging from 0.11 for TY to 0.27 for TM, all with $p < 0.001$. Deep-profile (D) collars had a small but significant negative effect on CH_4 flux (estimate = -0.03, $p = 0.006$) compared to B collars, while Surface (S) collars had no significant effect on CH_4 emissions (estimate = 0.01, $p = 0.58$). Post-hoc pairwise comparisons (Supporting Information) further confirmed that CH_4 flux was significantly lower in Palsas compared to all thermokarst locations ($p < 0.001$), with the largest difference observed for TM (estimate = -0.269, $p < 0.001$), followed by TR (estimate = -0.224, $p < 0.001$). CH_4 fluxes also exhibited significant differences among thermokarst sites, with TM showing significantly higher fluxes than TY (estimate = 0.158, $p < 0.001$) and TR exhibiting higher emissions than TY (estimate = 0.113, $p < 0.001$).

For CO_2 flux, the LMM results indicated that T and TM exhibited significantly higher fluxes compared to palsas, with estimates of 1.76 ($p < 0.001$) and 1.83 ($p < 0.001$), respectively. In contrast, TR did not show significant differences in CO_2 flux compared to palsas ($p = 0.29$), and TY exhibited a moderate increase (estimate = 0.95, $p = 0.01$). D collars significantly reduced CO_2 flux (estimate = -0.82, $p < 0.001$) compared to B collars. S collars, in contrast, exhibited a significant increase in CO_2 flux (estimate = 3.07, $p < 0.001$). Post-hoc pairwise comparisons (Supporting Information) further supported these trends, confirming significantly higher CO_2 fluxes in TM and T compared to Palsas, with estimates of -1.827 ($p < 0.001$) and -1.763 ($p < 0.001$), respectively. CO_2 flux was also significantly greater in TM compared to TR (estimate = 2.191, $p < 0.001$), and TR recorded significantly lower CO_2 flux than TY (estimate = -1.319, $p = 0.007$).

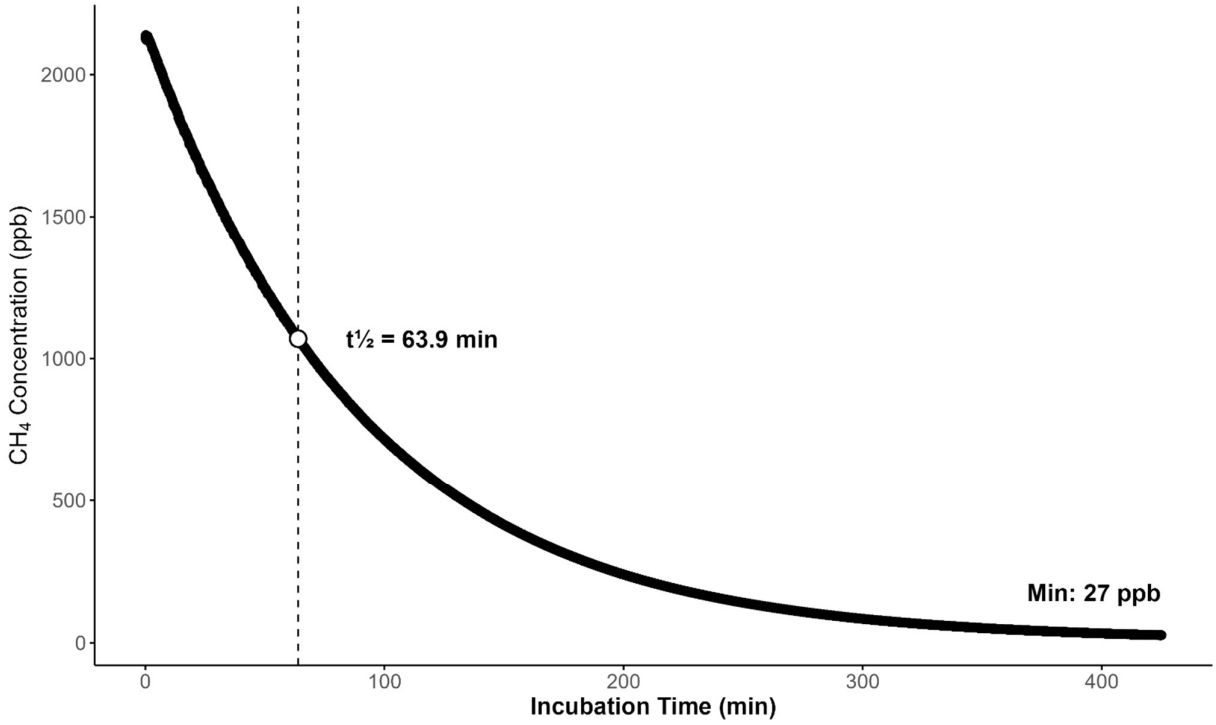


Figure 7. *CH_4 concentration over incubation time in palsa location APD3 from 11:55 – 19:00.*

CH_4 concentration in palsa locations was measured over a 425-minute incubation period (Figure 7). The initial CH_4 concentration was 2129 ppb, which rapidly declined within the first 100 minutes before slowing down. The half-life of CH_4 was 63.9 minutes, indicating the time required for CH_4 concentration to decrease by half. The decline continued progressively, reaching its lowest recorded concentration of 27 ppb at 424.9 minutes, after which no further substantial decrease was observed.

3.3 Effects of environmental variables on CH_4 and CO_2 fluxes

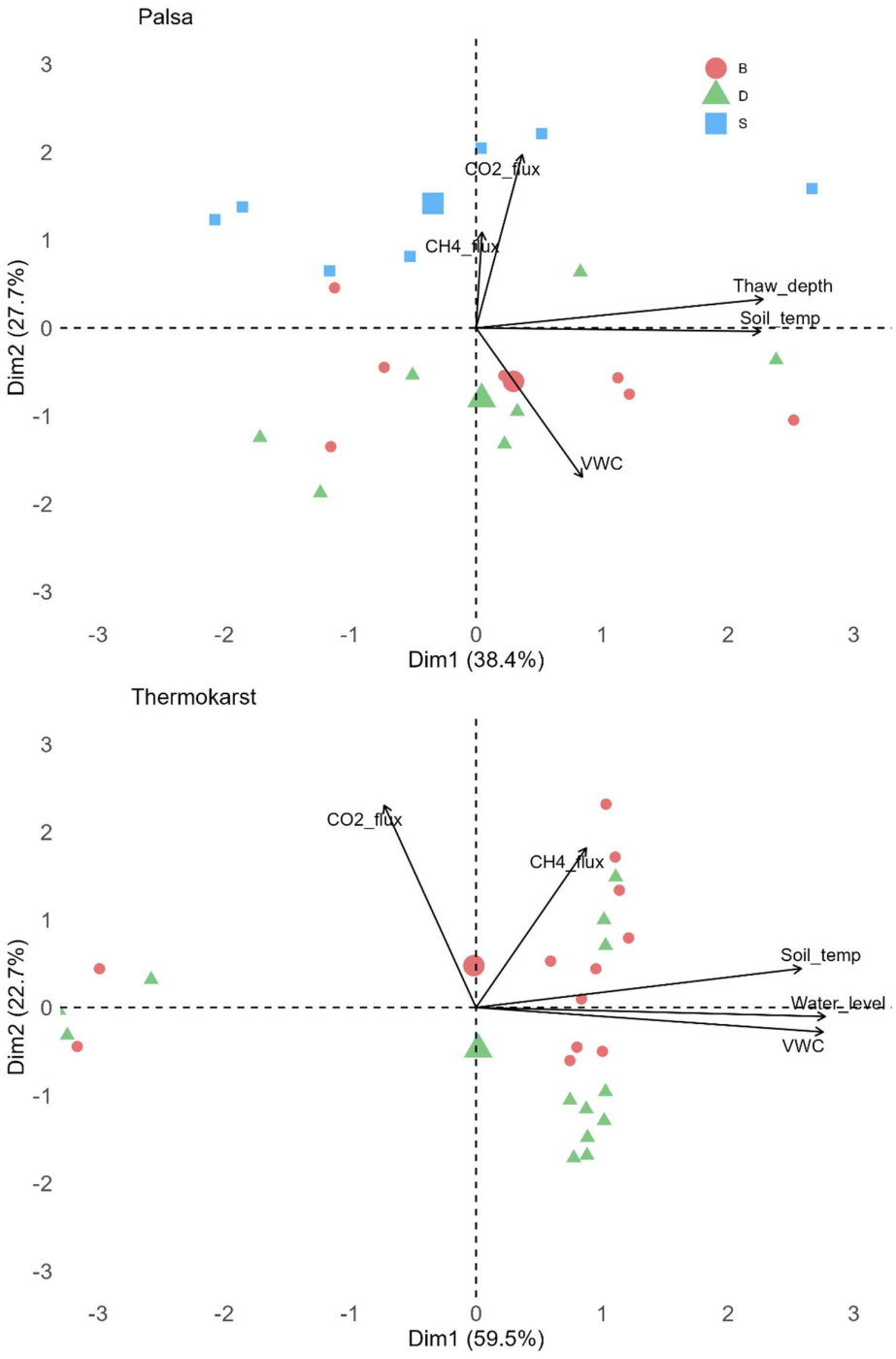


Figure 8. *Principal Component Analysis (PCA) biplots illustrating the relationships between environmental variables (soil temperature, VWC, water level, thaw depth) and greenhouse gas fluxes (CO_2 and CH_4) for palsas and thermokarsts. Dim1 and Dim2 represent the first two principal components with their explained variances. Symbols represent collar treatments: circles (B), triangles (D), and squares (S). Each value represents a seasonal average for each collar ID. Vectors indicate the contribution and direction of each variable to the principal components.*

The PCA results for Palsa and Thermokarst sites reveal distinct clustering patterns and variable associations (Figure 8). In the Palsa PCA, the first principal component (Dim1) explained 38.4% of the variance, while the second principal component (Dim2) accounted for 27.7%. In the Thermokarst PCA, Dim1 contributed 59.5% of the variance, while Dim2 explained 22.7%.

In palsa locations, CO_2 flux and CH_4 flux clustered together along Dim1. Some of the visible correlations from the PCA are present with thaw depth and soil temperature ($r = 0.156$, $p = 0.032$) with a Spearman correlation analysis confirming a weak positive correlation. Additionally, it can be deduced that VWC was negatively correlated with both thaw depth ($r = -0.145$, $p = 0.022$) and soil temperature ($r = -0.211$, $p = 0.019$), given the direction of the vectors. For the gas fluxes, CH_4 flux was significantly correlated with CO_2 flux ($r = 0.145$, $p = 0.046$) and soil temperature ($r = -0.173$, $p = 0.017$). While, despite a similar direction in the vectors, a negative correlation between thaw depth and CO_2 flux ($r = -0.181$, $p = 0.013$) was present.

In the Thermokarst PCA, CO_2 and CH_4 fluxes similarly mostly clustered along Dim1, while soil temperature, water level, and VWC were aligned along Dim2. Some relationships seen in the PCA were confirmed by the Spearman correlation analysis, including a strong positive correlation between soil temperature and VWC ($r = 0.551$, $p < 0.001$). Water level was also significantly correlated with soil temperature ($r = 0.265$, $p < 0.001$), as well as with VWC ($r = 0.450$, $p < 0.001$). Regarding gas fluxes, CH_4 flux showed a significant correlation with CO_2 flux ($r = 0.349$, $p < 0.001$) and was also correlated with soil temperature ($r = 0.165$, $p = 0.012$). A negative correlation between water level and CO_2 flux ($r = -0.374$, $p < 0.001$) was present, given the differing directionality in the vectors. Additionally, VWC was significantly correlated with CO_2 flux ($r = 0.134$, $p = 0.041$), despite a similar visible trend to water level. Finally, although water level and CH_4 flux did not show clear contrast in directionality in the PCA, they were found to be negatively correlated ($r = -0.137$, $p = 0.037$).

3.4 Comparison of collar treatments in palsas

Table 3-2. *Linear Mixed-Effects Model summaries for Box-Cox transformed CH_4 flux and CO_2 flux in palsas. Fixed effects include collar treatments (D, S, with B as the reference level). Estimates represent the direction and magnitude of the effect on gas flux.*

Predictor	Estimate	Std. Error	p-value
CH4			
(Intercept)	0.38	0.03	0.00***
TreatmentD	-0.12	0.05	0.01*
TreatmentS	0.00	0.05	0.97
CO2			
(Intercept)	3.50	0.06	0.00***
TreatmentD	-0.14	0.07	0.05
TreatmentS	0.45	0.07	0.00***

Note:
Significance levels: *** p < 0.001, ** p < 0.01, * p < 0.05. Estimates are derived from Box-Cox transformed flux values.

The influence of collar treatments on CH_4 and CO_2 fluxes in palsas was further assessed using linear mixed-effects models (Table 2) and post hoc pairwise comparisons (Supporting Information), which revealed significant treatment effects and variations across collar types. Deep-profile collars recorded a significant reduction in CH_4 flux compared to the reference B collars (estimate = -0.12 ± 0.03, p = 0.014). Post hoc pairwise comparisons (Supporting Information) further revealed that D collars exhibited significantly lower CH_4 flux than B collars (estimate = -0.12 ± 0.05, p = 0.040) and S collars (estimate = -0.12 ± 0.05, p = 0.033). However, S collars showed no significant difference from B collars (estimate = -0.002 ± 0.05, p = 0.999).

For CO_2 flux in palsas, collar treatment also influenced flux rates. S collars recorded a significantly higher CO_2 flux compared to B collars (estimate = -0.45 ± 0.07, p < 0.001). Post hoc analyses (Supporting Information) for CO_2 flux revealed that S collars exhibited significantly higher fluxes than both B collars (estimate = -0.45 ± 0.07, p < 0.001) and D collars (estimate = -0.60 ± 0.07, p < 0.001). However, D collars did not significantly differ from B collars (estimate = 0.14 ± 0.08, p = 0.132).

3.5 Chronosequence analysis of thermokarst development

Table 3-3. *Linear Mixed-Effects Model summaries for Box-Cox transformed CH_4 flux and CO_2 flux in thermokarst locations. The predictors include Treatment and thermokarst sublocations (TM, TR, TY, with T as the reference level). Estimates represent the direction and magnitude of the effect on gas flux.*

Predictor	Estimate	Std. Error	p-value
CH4			
(Intercept)	0.38	0.01	0.00***
LocationTM	0.11	0.01	0.00***
LocationTR	0.06	0.02	0.00***
LocationTY	-0.05	0.01	0.00***
TreatmentD	-0.01	0.01	0.28
CO2			
(Intercept)	10.77	0.54	0.00***
LocationTM	0.39	0.76	0.61
LocationTR	-1.79	0.74	0.02*
LocationTY	-0.48	0.75	0.52
TreatmentD	-1.03	0.23	0.00***

Note:
Significance levels: *** p < 0.001, ** p < 0.01, * p < 0.05. Estimates are derived from Box-Cox transformed flux values.

The linear mixed-effects model identified a significant effect of location on CH_4 flux, with TM, TR, and TY differing significantly from T (LocationTM: 0.11 ± 0.01, $p < 0.001$; LocationTR: 0.06 ± 0.02, $p < 0.001$; LocationTY: -0.05 ± 0.01, $p < 0.001$). No significant effect of treatment was detected (TreatmentD: -0.01 ± 0.01, $p = 0.28$). Post-hoc comparisons (Supporting Information) revealed that CH_4 flux at T was significantly lower than at TM (Estimate = -0.106 ± 0.013, $p < 0.001$) and TR (Estimate = -0.064 ± 0.016, $p < 0.001$) but significantly higher than TY (Estimate = 0.052 ± 0.010, $p < 0.001$). TM exhibited significantly higher CH_4 flux than TR (Estimate = 0.042 ± 0.015, $p = 0.032$) and TY (Estimate = 0.158 ± 0.010, $p < 0.001$). CH_4 flux at TR was significantly lower than TY (Estimate = 0.116 ± 0.013, $p < 0.001$).

Similarly, CO_2 flux in thermokarst locations also exhibited significant variation based on location. The linear mixed-effects model showed that location strongly influenced CO_2 flux rate with TR exhibiting lower flux than the reference (Estimate = -1.79 ± 0.74, $p = 0.02$). Treatment also had a significant effect (TreatmentD: -1.03 ± 0.23, $p < 0.001$). Post-hoc analyses (Supporting Information) further showed that TM had significantly higher CO_2 flux than TR (Estimate = 2.176 ± 0.334, $p < 0.001$), while

TR was significantly lower than TY (Estimate = -1.310 ± 0.321, $p < 0.001$). Flux differences between other location pairs, such as T and TM, or T and TY, were not significant.

Table 3-4. *Compilation of literature values for CH4 and CO2 fluxes across different permafrost landforms and locations. * Excludes 3 measurements for improved comparison of the seasonal range of fluxes. [a] Standard deviation. [b] Standard error.*

Permafrost landform	***CO_2 Flux ($g\ m^{-2}\ d^{-1}$)***	***CH_4 Flux ($mg\ m^{-2}\ d^{-1}$)***	***Location***	***Source***
Palsa	1.28 to 17.61	-0.06 to −8.28* 6.08 to −8.28	Finnmark, Norway	This study
Thermokarst	0.12 to 12.93	0.06 to 2293* −3.35 to 2293	Finnmark, Norway	This study
Palsa	4.78 ± 2.16[a]		Western Siberia	(Matyshak et al., 2023)
Thermokarst bog	4.37 ± 2.04[a]		Western Siberia	(Matyshak et al., 2023)
Thermokarst bog	5.02	80	Western Canada	(Heffernan et al., 2024)
Peat plateau	4.91 ± 0.44		Western Canada	(Heffernan et al., 2024)
Palsa (ridge/peat mound)		6.72 to 20.4	Northeastern European Russia	(Zagirova et al., 2023)
Palsa mire		−11.07 to 18.13	North-central Siberia	(Olchev et al., 2022)
Permafrost mineral soil		−0.10	Northern Siberia	(Flessa et al., 2008)
Thermokarst pond		54.79	Northern Siberia	(Flessa et al., 2008)
Palsa mire (Lichen communities)		−1.44	Northeastern European Russia	(Miglovets et al., 2021)
Palsa mire (Peat spots)		0.96	Northeastern European Russia	(Miglovets et al., 2021)
Palsa		−3.43	Finnish Lapland	(Voigt et al., 2023)
Dry tundra		−3.19 ± 1.42	Northeast Greenland	(Juncher Jørgensen et al., 2015)
Bare soil permafrost		−1.92 ± 0.48 to −4.80 ± 0.96[b]	West Greenland	(D'Imperio et al., 2017)
Palsa-bog complex	3.60 to 6.12		Western Siberia	(Goncharova et al., 2023)
Peat plateau	9.58 ± 6.56[a]	6.14 ± 10.75[a]	Alaska	(Wickland et al., 2006)
Thermokarst wetland	9.77 ± 6.56[a]	172.8 ± 126.72[a]	Alaska	(Wickland et al., 2006)
Thermokarst lakes		33.6 ± 4.8[b]	Tibetan Plateau	(Yang et al., 2023)
Thermokarst lake		1,170	Alaska	(Elder et al., 2021)
Thermokarst lakes		0.048 to 774.4	Tibetan Plateau	(L. Wang et al., 2021)

4 Discussion

4.1 CO_2 and CH_4 flux differences between palsas and thermokarst

The results confirmed significant differences in CH_4 and CO_2 fluxes between palsas and thermokarst sites. The LMM results indicated that CH_4 flux was consistently higher in thermokarst features compared to palsas, supporting the hypothesis that anaerobic conditions in water-saturated thermokarst soils promote methanogenesis. This matches findings from other studies demonstrating that permafrost thaw increases CH_4 production under anoxic conditions (Heffernan et al., 2022). Post hoc comparisons confirmed that TM exhibited the highest CH_4 flux, while TY and TR also showed significantly greater CH_4 emissions than palsas. However, Heffernan et al. (2022) reported that CH_4 fluxes were highest in younger thermokarst bogs (approximately 30 years post-thaw) and decreased in mature bogs (around 200 years post-thaw). This discrepancy suggests that while this study observed higher CH_4 emissions in more mature thermokarst features, other research indicates that CH_4 fluxes may peak during earlier successional stages and decline as the bog matures. One possible explanation for the higher CH_4 emissions observed in mature thermokarst features in our study is the accumulation of organic matter over time, which provides a continuous substrate for methanogenic archaea under anoxic conditions, leading to sustained or even increased methane production (Kallistova et al., 2019). The observed variability in CH_4 flux was closely linked to soil moisture conditions, as indicated by the strong correlation between CH_4 flux and VWC, consistent with other studies which found that CH_4 fluxes in permafrost peatlands were strongly regulated by water table depth and soil moisture content (Natali et al., 2015; Turetsky et al., 2008). The time series analysis showed stable water levels in thermokarst sites, reinforcing the sustained anaerobic conditions that promote methane production, while CH_4 flux fluctuations coincided with increases in VWC and soil temperature.

The observed CH_4 fluxes in thermokarst features are comparable to those reported in similar environments. For instance, Heffernan et al. (2024) recorded CH_4 fluxes of 80 mg m^{-2} d^{-1} in a thermokarst bog in Western Canada, aligning with the flux range observed in this study. Similarly, studies in Western Siberia have reported CH_4 emissions from thermokarst bogs of comparable magnitudes (Matyshak et al., 2023). The high CH_4 fluxes in these environments are attributed to prolonged water saturation, which maintains anoxic conditions conducive to methanogenesis (Zandt et al., 2020). Additionally, organic matter availability plays a crucial role, as seen in other works, where CH_4 fluxes in thermokarst lakes were significantly influenced by substrate quality and microbial activity (Heslop et al., 2020; in 't Zandt et al., 2020).

Conversely, CO_2 flux exhibited substantial variability across sites, with T and TM recording significantly higher fluxes compared to palsas, while TY exhibited moderately higher CO_2 flux, and TR did not differ significantly from Palsas. This pattern partially supports the hypothesis that lower VWC and progressive thaw depth in palsas create aerobic conditions that enhance microbial decomposition (Hodgkins et al., 2014; Natali et al., 2015). The time series analysis revealed a seasonal increase in thaw depth in palsas, which may have influenced the release of previously frozen carbon stocks, contributing to sustained CO_2 emissions. However, the fact that T and TM exhibited significantly greater CO_2 fluxes than palsas suggests that additional factors, such as increased vegetation productivity and microbial activity, may sustain high CO_2 emissions in these sites. As permafrost thaws, previously frozen organic matter becomes available, leading to enhanced microbial decomposition and subsequent CO_2 release. Additionally, elevated temperatures and CO_2 levels can stimulate plant biomass production, further contributing to increased respiration and carbon fluxes (Ruan et al., 2023). In contrast, TR exhibited the lowest CO_2 emissions among thermokarst sites, likely due to its early-stage thaw conditions and limited vegetation development, which may have constrained microbial decomposition. Early-stage thaw sites may not yet have accumulated sufficient organic matter or developed extensive plant communities to support high levels of microbial activity (Zolkos et al., 2019). Thermokarst locations also exhibited relatively stable water levels, which limit oxygen diffusion into the soil, thereby creating anaerobic conditions that constrain aerobic microbial respiration. This is supported by the significant negative correlation observed between CO_2 flux and water level, indicating that higher water levels are associated with lower CO_2 emissions due to restricted aerobic decomposition (Zolkos et al., 2019).

CO_2 fluxes in this study are within the range reported in previous research. For example, Matyshak et al. (2023) observed CO_2 fluxes of 4.78 ± 2.16 g m^{-2} d^{-1} in Western Siberian palsas, which aligns with the fluxes measured in this study. Similarly, Heffernan et al. (2024) reported CO_2 fluxes of 5.02 g m^{-2} d^{-1} in a thermokarst bog, which is comparable to the values observed in TM and TY. These studies indicate that while thermokarst features generally emit lower CO_2 than palsas, increased vegetation productivity in certain thermokarst sites can sustain CO_2 emissions at levels similar to or higher than palsas.

One of the key findings of this study was the presence of negative CH_4 fluxes (uptake) in palsa sites, suggesting net CH_4 oxidation. Similar patterns have been reported in other studies, such as Voigt et al. (2023), who observed CH_4 uptake of -3.43 mg m^{-2} d^{-1} in Finnish Lapland palsas. Other studies in Northeast Greenland (Juncher Jørgensen et al., 2015) and West Greenland (D'Imperio et al., 2017) have also documented CH_4 uptake in dry tundra and permafrost mineral soils. The negative CH_4 fluxes observed in this study are likely driven by well-aerated soil conditions that favor methanotrophic

bacterial activity (Fest et al., 2017). In contrast to thermokarst features, where CH_4 production dominates under anoxic conditions, palsas provide an environment where methane oxidation can overcome methane production, leading to net uptake. Additionally, vegetation cover, particularly the presence of lichen and mosses, can enhance CH_4 oxidation by influencing soil microclimate and microbial community composition (Belova et al., 2020).

4.2 The influence of environmental variables on CH_4 and CO_2 fluxes

The results of this study partially support the hypothesis that higher soil temperatures enhance greenhouse gas fluxes, though the effects differed between CO_2 and CH_4. While CH_4 flux showed a significant positive correlation with soil temperature in thermokarsts, CO_2 flux did not exhibit a direct temperature dependency, instead being more strongly influenced by hydrological conditions.

The observed positive correlation between CH_4 flux and soil temperature in thermokarst sites suggests that warmer conditions may enhance methanogenic activity, particularly in waterlogged soils. Previous studies have shown that CH_4 emissions tend to increase with rising temperatures, as microbial decomposition accelerates under anoxic conditions (Knoblauch et al., 2018, 2018; Olefeldt et al., 2013). This aligns with findings from de Jong et al. (2018), who reported that CH_4 production in thermokarst lakes was enhanced at higher temperatures, likely due to increased methanogenic substrate availability and microbial activity. However, in this study, the magnitude of this effect was more moderate, suggesting that temperature alone may not be the dominant driver of CH_4 flux variability in these sites. Other interacting factors, such as substrate limitation, hydrological conditions, or microbial community composition, may also regulate methane production in response to warming. For instance, research indicates that while temperature is a key factor in peat decomposition, other constraints like oxygen availability and high concentrations of phenolic compounds can limit microbial activity and methane production (Kluber et al., 2020). Additionally, studies have shown that soil and methane-cycling microbial community structures are strongly influenced by water level changes and saturation duration, affecting methane emissions (Maietta et al., 2020). Therefore, a combination of thermal and environmental factors likely governs CH_4 flux dynamics in thermokarst landscapes.

In contrast to CH_4 flux, CO_2 flux did not significantly correlate with soil temperature in either palsa or thermokarst sites. Instead, CO_2 flux was negatively correlated with water level in thermokarsts, suggesting that hydrological conditions, rather than temperature, play a larger role in regulating CO_2 emissions in these environments. This aligns with studies showing that water saturation in permafrost peatlands can constrain microbial activity, with higher water levels limiting oxygen diffusion and reducing aerobic microbial respiration (Beckebanze et al., 2022; Kwon et al., 2022). The observed negative correlation between water level and CO_2 flux supports the idea that increased soil saturation

suppresses aerobic microbial respiration, limiting CO_2 production. This trend has been widely reported in thermokarst ecosystems, where CO_2 fluxes tend to decrease as soil moisture increases due to reduced oxygen availability (Heffernan et al., 2024; Olefeldt et al., 2013).

While soil temperature did not strongly influence CO_2 flux, it was positively correlated with water level and VWC. This suggests that warming contributes to increased soil moisture availability, which in turn may indirectly influence CH_4 emissions. Similar patterns have been observed in Arctic permafrost peatlands, where rising temperatures are associated with enhanced permafrost thaw, leading to higher soil moisture and subsequent increases in CH_4 flux (Matveev et al., 2016).

These results emphasize the complex interactions between soil temperature, hydrology, and greenhouse gas fluxes in permafrost peatlands. While higher soil temperatures contributed to increased CH_4 emissions in thermokarsts, CO_2 fluxes were more constrained by hydrological conditions than by temperature. This suggests that future greenhouse gas emissions from thawing permafrost could likely depend not only on warming rates but also on water table dynamics and site-specific differences in soil moisture retention. Studies have suggested that as permafrost thaw progresses, increasing water saturation in thermokarst environments could lead to a shift in carbon dynamics, with declining CO_2 emissions and rising CH_4 production (Estop-Aragonés et al., 2020; Voigt et al., 2023). The findings of this study support this hypothesis, highlighting the importance of both thermal and hydrological controls on carbon cycling in degrading permafrost landscapes.

4.3 The influence of collar treatments on CH_4 and CO_2 fluxes

Collar treatments significantly influenced both CO_2 and CH_4 fluxes in palsas, highlighting the role of root respiration, soil aeration, and microbial activity in regulating greenhouse gas dynamics. The LMM results indicated that S collars exhibited significantly higher CO_2 flux compared to B collars and D collars, suggesting a strong contribution of autotrophic respiration from vegetation. The increase in CO_2 flux in S collars is consistent with studies showing that vascular plant root respiration and exudation can enhance CO_2 release from permafrost soils (Estop-Aragonés, Cooper, et al., 2018; Estop-Aragonés, Czimczik, et al., 2018; Lupascu et al., 2014). The vegetation present in palsas, primarily *Betula nana*, *Ledum palustre*, *Empetrum nigrum*, and *Vaccinium vitis-idaea*, likely contributed to this trend by increasing organic inputs and stimulating microbial decomposition of soil organic carbon. This aligns with Bret-Harte et al. (2001), who demonstrated that dwarf shrubs, such as *Betula nana*, can significantly influence carbon dynamics in Arctic tundra by increasing soil organic matter inputs and promoting microbial decomposition.

In contrast, D collars did not significantly increase CO_2 flux compared to B collars, despite their greater exposure to deeper soil layers. This was unexpected, as D collars were hypothesized to produce

higher CO_2 fluxes due to increased access to previously frozen organic carbon. However, the lack of a significant effect suggests that CO_2 production in deeper soil layers is not as immediate as surface respiration and may require longer timescales to detect. Similar findings have been reported in permafrost ecosystems, where greater subsurface exposure did not necessarily translate to increased CO_2 emissions (Elberling et al., 2013). This could be due to the relatively slow decomposition of deep organic matter, which may require different microbial communities or more pronounced warming before significant CO_2 release occurs (Kirkwood et al., 2021; Koven et al., 2015). Additionally, oxygen availability declines with depth, potentially limiting aerobic microbial respiration in D collars (Treat et al., 2015). If seasonal differences in CO_2 flux exist, it is possible that D collars require longer monitoring periods to detect their impact. Studies on deep permafrost carbon mobilization suggest that older, more recalcitrant carbon pools are released more slowly compared to fresh organic inputs from surface vegetation (Schuur et al., 2015). Future research focusing on seasonal and long-term trends could help determine whether D collars contribute more significantly to CO_2 fluxes over time rather than in short-term chamber-based measurements.

Regarding CH_4 flux, D collars exhibited the strongest net CH_4 uptake, while S collars showed no significant difference from B collars. This suggests that deeper soils in palsas may provide conditions more favorable for CH_4 oxidation, possibly due to the presence of methanotrophic microbial communities. These results align with Voigt et al. (2023), who reported that palsa soils frequently act as net methane sinks, particularly in well-aerated, dry environments where methane oxidation outpaces production. Similarly, Miglovets et al. (2021) demonstrated that CH_4 uptake is enhanced in palsa sites dominated by lichen and mosses, reinforcing the idea that palsa vegetation contributes to CH_4 oxidation. The fact that D collars exhibited the most negative CH_4 flux values suggests that they may have enhanced methane oxidation by providing an extended aerobic layer for methanotrophic bacteria to consume CH_4 before it reached the atmosphere (Kirkwood et al., 2021; Perryman et al., 2020).

Soil moisture may have also contributed to the strong CH_4 uptake in D collars. Drier conditions in palsa organic soils have been shown to increase methane oxidation rates due to greater oxygen availability. This reinforces findings from methane oxidation studies in permafrost peatlands, where dry aerobic soil layers act as CH_4 sinks, while wetter conditions suppress oxidation (Voigt et al., 2023). Their study demonstrated that soil drying, and increased nutrient supply promote CH_4 uptake in Arctic soils, emphasizing the role of oxygen availability in enhancing methane oxidation processes.

Interestingly, B collars exhibited greater CH_4 uptake than expected, despite their sealed bottom restricting deeper gas exchange. One possible explanation is that CH_4 oxidation is occurring primarily

in the upper soil layers, where oxygen is more available. Studies in permafrost peatlands have shown that even a thin aerobic zone near the surface can oxidize a substantial fraction of CH_4 before it reaches the atmosphere (Voigt et al., 2023). Additionally, while B collars were designed to limit contributions from deeper soil layers, it is possible that small drainage holes allowed some gas diffusion, which may have enabled continued microbial activity and oxidation. However, this was not a design feature intended to influence CH_4 flux measurements, so further investigation would be required to confirm its effect.

Another key observation was that S collars did not significantly differ from B collars in terms of CH_4 flux. Unlike sedge-dominated thermokarst wetlands, where vascular plants transport CH_4 through aerenchyma tissues (Bhullar et al., 2013; Riutta et al., 2020; Ström et al., 2012), the vascular plant species in palsas—mainly dwarf shrubs—do not facilitate methane transport to the same extent. This aligns with studies showing that shrublike vegetation promotes CH_4 oxidation rather than direct transport, likely due to increased root oxygenation (Määttä & Malhotra, 2024; Voigt et al., 2023). Thus, the lack of a CH_4 increase in S collars compared to B collars suggests that vascular plant effects were more influential on CO_2 flux rather than CH_4 emissions.

4.4 CH_4 and CO_2 flux differences across thermokarst stages

Thermokarst formation and subsequent ecosystem succession create dynamic changes in hydrology, soil properties, and vegetation, all of which influence greenhouse gas emissions. The results of this study indicate that CH_4 and CO_2 fluxes vary significantly across different stages of thermokarst development, reflecting the interplay between microbial activity, vegetation succession, and environmental conditions.

Methane (CH_4) emissions exhibit variability across different stages of thermokarst development, rather than following a straightforward increasing or decreasing pattern. Notably, the highest CH_4 emissions are often observed in mature thermokarst features, which provide stable, water-saturated conditions that promote methanogenesis. This is consistent with findings from studies on alpine thermokarst lakes on the Tibetan Plateau, which have shown that such lakes emit significant amounts of methane due to the presence of methanogenic microorganisms thriving in anaerobic, waterlogged environments (Yang et al., 2023). Similarly, research on different stages of thermokarst formation in subarctic regions has demonstrated that younger thermokarst stages with shallow ponds can exhibit higher CH_4 emissions from the pond surface during warmer periods, highlighting the complex interplay between thermokarst development stages and methane emissions (Desyatkin et al., 2009). The persistence of standing water in these environments, combined with the accumulation of decomposable organic matter, facilitates anaerobic microbial activity and methane production. This

underscores the importance of hydrological conditions and organic matter availability in regulating methane emissions across various thermokarst stages.

In contrast, CH_4 fluxes in TR were significantly lower than in TM but still greater than in T and TY. This suggests that while recently thawed sites release methane due to the exposure of previously frozen organic carbon, they may not yet support the fully developed hydrological conditions required for sustained methanogenesis. Interestingly, significant increases in CH_4 fluxes were observed around Julian Day 220, particularly for T and TR sites. This increase was particularly evident around August 14th (Julian Day 226), where TR exhibited some of the highest CH_4 fluxes of the study period (252.18 mg m^{-2} d^{-1}, 108.99 mg m^{-2} d^{-1}, and 2293.38 mg m^{-2} d^{-1}), and T also recorded higher-than-usual values (124.17 mg/m²/day). The increase in CH_4 fluxes can likely be attributed to the combination of warmer temperatures and the effect of drought, which promoted microbial activity and higher productivity in the sub peat. These fluxes are more comparable to those found in thermokarst lake in Alaska by Elder et al. (2021).

The lowest CH_4 fluxes were observed in TY, which was significantly lower than all other thermokarst stages. This aligns with findings from other studies which reported that increased vegetation cover, particularly with colonizing mosses and shrubs, can lead to a net reduction in CH_4 emissions due to enhanced oxygenation of the rhizosphere and increased methane oxidation (Liebner et al., 2011; Murase et al., 2020). T showed intermediate CH_4 emissions, lower than TR and TM but higher than TY, which suggests that initial thaw conditions create an environment where CH_4 is still being produced, though not at the levels seen in fully developed thermokarst systems.

CO_2 fluxes exhibited a different trend, with TM recording the highest emissions, followed by T and TY, while TR had the lowest CO_2 fluxes. The significantly greater emissions in TM compared to TR align with studies that highlight the role of vegetation and microbial decomposition in driving CO_2 fluxes in permafrost peatlands (Serikova et al., 2019). Mature thermokarst sites tend to develop a dense vegetation cover, which contributes to enhanced autotrophic respiration through increased root biomass and exudates (Hicks Pries et al., 2013). Additionally, increased decomposition of organic matter in the surface layer further promotes CO_2 release (Lee et al., 2010). In contrast, TR exhibited the lowest CO_2 emissions, possibly due to limited vegetation and relatively fresh thaw conditions, where microbial activity has not yet fully mobilized carbon stores.

T exhibited comparable CO_2 fluxes to TM, despite being an earlier stage in thermokarst development. One possible explanation is that initial thaw conditions lead to high rates of microbial decomposition as previously frozen organic matter is exposed, temporarily increasing CO_2 emissions before

stabilization occurs (G. Wang et al., 2024). In TY, CO_2 emissions were slightly lower than in T but remained higher than in TR, suggesting that vegetation succession may have contributed to stabilizing carbon loss while maintaining respiration-driven fluxes (Skeeter et al., 2020). The post hoc analysis revealed that while TM had significantly greater CO_2 fluxes than TR, no significant differences were detected between T and other thermokarst stages, highlighting the variability in CO_2 dynamics across early thermokarst development.

5 Conclusion

This study provides new insights into the effects of permafrost thaw on greenhouse gas fluxes in Arctic peatlands, specifically examining CH_4 and CO_2 flux dynamics across palsas and thermokarst features in Finnmark, Norway. By integrating in-situ chamber measurements with environmental variables, the study demonstrates that permafrost degradation significantly alters carbon cycling, with thermokarst sites acting as CH_4 sources and palsas functioning as CH_4 sinks. CO_2 fluxes were found to be more strongly influenced by hydrological conditions than temperature, highlighting the importance of water table dynamics in regulating carbon emissions from thawing permafrost.

Key findings indicate that CH_4 fluxes were highest in thermokarsts, particularly in TM, where stable waterlogged conditions favored methanogenesis, while palsas maintained net CH_4 uptake. CH_4 emissions correlated positively with soil temperature in thermokarsts but showed no significant relationship in palsas, suggesting that moisture availability is a stronger limiting factor in drier permafrost soils. CO_2 fluxes exhibited a distinct negative correlation with water level, reinforcing the role of oxygen availability in controlling microbial respiration. These results emphasize that temperature alone does not drive permafrost carbon emissions, but rather a complex interplay between warming, soil moisture, and vegetation dynamics. The study also highlights the importance of collar design in gas flux measurements, with deeper-profile collars facilitating greater CH_4 oxidation due to increased oxygen penetration. Thermokarst development stages exhibited nonlinear flux trends, indicating that methane and carbon dioxide emissions are highly dependent on site-specific hydrology, vegetation succession, and microbial activity. As permafrost thaw accelerates under climate change, the future trajectory of Arctic and subarctic carbon emissions will depend on shifts in hydrology and ecosystem succession. While thermokarst formation enhances CH_4 emissions in waterlogged conditions, prolonged drying of thawed peatlands may favor CO_2 release through increased microbial decomposition. Future work should focus on long-term monitoring of thermokarst succession, integrating remote sensing data with field-based flux measurements to refine projections of permafrost carbon loss in a warming climate. Understanding the balance between CH_4 and CO_2 emissions in permafrost peatlands remains critical for assessing their long-term impact on global climate change.

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Table 8-1. *Significant post hoc pairwise comparisons of CH_4 flux between landform types using Tukey's adjustment. "Comparison" indicates the two groups being compared, "Estimate" represents the mean difference in CH_4 flux between them, "Std. Error" is the standard error of the estimate, "df" is the degrees of freedom, "t.ratio" is the test statistic, and "p-value" indicates the significance level. Negative estimates indicate lower CH_4 flux in the first group compared to the second.*

Significant Post-Hoc Comparisons for CH4 Flux

Comparison	Estimate	Std. Error	df	t.ratio	p-value
P - T	-0.163	0.018	403	-9.155	0.000
P - TM	-0.269	0.018	403	-15.182	0.000
P - TR	-0.224	0.019	403	-11.883	0.000
P - TY	-0.111	0.016	403	-6.819	0.000
T - TM	-0.106	0.021	403	-4.975	0.000
T - TR	-0.061	0.022	403	-2.748	0.049
TM - TY	0.158	0.020	403	7.918	0.000
TR - TY	0.113	0.021	403	5.395	0.000

Note:

Only significant comparisons ($p < 0.05$) are shown.

Table 8-2. *Significant post hoc pairwise comparisons of CO_2 flux between landform types using Tukey's adjustment. "Comparison" indicates the two groups being compared, "Estimate" represents the mean difference in CH_4 flux between them, "Std. Error" is the standard error of the estimate, "df" is the degrees of freedom, "t.ratio" is the test statistic, and "p-value" indicates the significance level. Negative estimates indicate lower CH_4 flux in the first group compared to the second.*

Significant Post-Hoc Comparisons for CO2 Flux

Comparison	Estimate	Std. Error	df	t.ratio	p-value
P - T	-1.763	0.321	407	-5.489	0.000
P - TM	-1.827	0.304	407	-6.010	0.000
T - TR	2.128	0.423	407	5.034	0.000
TM - TR	2.191	0.320	407	6.842	0.000
TR - TY	-1.319	0.391	407	-3.370	0.007

Note:

Only significant comparisons ($p < 0.05$) are shown.

Table 8-3. *Post hoc pairwise comparisons of CH_4 flux between Collar treatment using Tukey's adjustment. "Comparison" indicates the two groups being compared, "Estimate" represents the mean difference in CH_4 flux between them, "Std. Error" is the standard error of the estimate, "df" is the degrees of freedom, "t.ratio" is the test statistic, and "p-value" indicates the significance level. Negative estimates indicate lower CH_4 flux in the first group compared to the second.*

contrast	estimate	SE	df	t.ratio	p.value
B - D	0.115	0.047	177	2.454	0.040
B - S	-0.002	0.046	177	-0.040	0.999
D - S	-0.117	0.046	177	-2.529	0.033

Note:

Post hoc pairwise comparisons for CH4 using Tukey's adjustment.

Table 8-4. *Post hoc pairwise comparisons of CO_2 flux between Collar treatment using Tukey's adjustment. "Comparison" indicates the two groups being compared, "Estimate" represents the mean difference in CH_4 flux between them, "Std. Error" is the standard error of the estimate, "df" is the degrees of freedom, "t.ratio" is the test statistic, and "p-value" indicates the significance level. Negative estimates indicate lower CO_2 flux in the first group compared to the second.*

contrast	estimate	SE	df	t.ratio	p.value
B - D	0.144	0.075	181	1.936	0.132
B - S	-0.452	0.073	181	-6.177	0.000
D - S	-0.596	0.073	181	-8.166	0.000

Note:

Post hoc pairwise comparisons for CO2 using Tukey's adjustment.

Table 8-5. *Significant post hoc pairwise comparisons of CH_4 flux between landform types using Tukey's adjustment. "Comparison" indicates the two groups being compared, "Estimate" represents the mean difference in CH_4 flux between them, "Std. Error" is the standard error of the estimate, "df" is the degrees of freedom, "t.ratio" is the test statistic, and "p-value" indicates the significance level. Negative estimates indicate lower CH_4 flux in the first group compared to the second.*

contrast	estimate	SE	df	t.ratio	p.value
T - TM	-0.106	0.013	222	-8.350	0.000
T - TR	-0.064	0.016	222	-4.111	0.000
T - TY	0.052	0.010	222	5.075	0.000
TM - TR	0.042	0.015	222	2.751	0.032
TM - TY	0.158	0.010	222	15.674	0.000
TR - TY	0.116	0.013	222	8.581	0.000

Note:

Post hoc pairwise comparisons using Tukey's adjustment for CH4 flux.

Table 8-6. *Significant post hoc pairwise comparisons of CO_2 flux between landform types using Tukey's adjustment. "Comparison" indicates the two groups being compared, "Estimate" represents the mean difference in CH_4 flux between them, "Std. Error" is the standard error of the estimate, "df" is the degrees of freedom, "t.ratio" is the test statistic, and "p-value" indicates the significance level. Negative estimates indicate lower CH_4 flux in the first group compared to the second.*

contrast	estimate	SE	df	t.ratio	p.value
T - TM	-0.387	0.756	222	-0.511	0.956
T - TR	1.790	0.743	222	2.410	0.078
T - TY	0.480	0.751	222	0.639	0.919
TM - TR	2.176	0.334	222	6.517	0.000
TM - TY	0.867	0.353	222	2.457	0.070
TR - TY	-1.310	0.321	222	-4.075	0.000

Note:

Post hoc pairwise comparisons using Tukey's adjustment for CO2 flux.